

CCNA Complete Flashcard Handbook & Reviewer

Credit note

This set of flashcards is from Jeremy's IT Lab. This reviewer is intended to help make review of the key takeaways from his CCNA lessons more efficient. Full credit for the original flashcard content goes to Jeremy's IT Lab.

Coverage summary

Section	Cards
Day 01 Flashcards - Network Devices	22
Day 02 Flashcards - Interfaces and Cables	57
Day 03 Flashcards - OSI Model, TCP/IP Suite	41
Day 04 Flashcards - Intro to the CLI	28
Day 05 Flashcards - Ethernet LAN Switching (Part 1)	38
Day 06 Flashcards - Ethernet LAN Switching (Part 2)	17
Day 07 Flashcards - IPv4 Addresses (Part 1)	27
Day 08 Flashcards - IPv4 Addresses (Part 2)	17
Day 09 Flashcards - Switch Interfaces	23
Day 10 Flashcards - IPv4 Header	56
Day 11 (part 1) Flashcards - Routing Fundamentals	16
Day 11 (part 2) Flashcards - Static Routing	13
Day 13 Flashcards - Subnetting	21
Day 16 Flashcards - VLANs (Part 1)	12
Day 17 Flashcards - VLANs (Part 2)	33
Day 18 Flashcards - VLANs (Part 3)	9
Day 19 Flashcards - DTP & VTP	44
Day 20 Flashcards - STP (Part 1)	36
Day 21 Flashcards - STP (Part 2)	41
Day 21-1 Flashcards - PortFast	8
Day 21-2 Flashcards - BPDU Guard & BPDU Filter	21
Day 21-3 Flashcards - Root Guard	6
Day 21-4 Flashcards - Loop Guard	12
Day 22 Flashcards - Rapid STP	39

Section	Cards
Day 23 Flashcards - EtherChannel	18
Day 24 Flashcards - Dynamic Routing	51
Day 25 Flashcards (1) - RIP & EIGRP	31
Day 25 Flashcards (2 - Lab Video) - EIGRP Terms	12
Day 26 Flashcards - OSPF (Part 1)	34
Day 27 Flashcards - OSPF (Part 2)	47
Day 28 Flashcards - OSPF (Part 3)	40
Day 29 Flashcards - FHRPs	36
Day 30 Flashcards - TCP & UDP	67
Day 31 Flashcards - IPv6 (Part 1)	13
Day 32 Flashcards - IPv6 (Part 2)	75
Day 33 Flashcards - IPv6 (Part 3)	45
Day 34 Flashcards - Standard ACLs	22
Day 35 Flashcards - Extended ACLs	26
Day 36 Flashcards - CDP & LLDP	40
Day 37 Flashcards - NTP	39
Day 38 Flashcards - DNS	20

Day 39 Flashcards - DHCP	26
Day 40 Flashcards - SNMP	33
Day 41 Flashcards - Syslog	40
Day 42 Flashcards - SSH	28
Day 43 Flashcards - FTP&TFTP	36
Day 44 Flashcards - NAT (Part 1)	19
Day 45 Flashcards - NAT (Part 2)	14
Day 46 Flashcards - QoS (Part 1)	41
Day 47 Flashcards - QoS (Part 2)	55
Day 48 Flashcards - Security Fundamentals	54
Day 49 Flashcards - Port Security	34
Day 50 Flashcards - DHCP Snooping	21
Day 51 Flashcards - Dynamic ARP Inspection	17
Day 52 Flashcards - LAN Architectures	21
Day 53 Flashcards - WAN Architectures	37
Day 54 (Part 1) Flashcards - Virtualization & Cloud	32
Day 54 (part 2) Flashcards - Containers	13
Day 54 (Part 3) Flashcards - VRF	9
Day 55 Flashcards - Wireless Fundamentals	69
Day 56 Flashcards - Wireless Architectures	57
Day 57 Flashcards - Wireless Security	43
Day 58 Flashcards - Wireless Configuration	34
Day 59 (Part 1) Flashcards - Network Automation	22
Day 59 (part 2) Flashcards - AI & Machine Learning	16
Day 60 Flashcards - JSON, XML, & YAML	26
Day 61 (Part 1) Flashcards - REST APIs	38
Day 61 (Part 2) Flashcards - REST Authentication	12
Day 62 Flashcards - SDN	19
Day 63 Flashcards - Ansible, Puppet, Chef	27
Day 63 (part 2) Flashcards - Terraform	27

Complete Card-by-Card Reviewer

The following section includes every question/flashcard from the uploaded deck, organized by its original CCNA day/subdeck.

Day 01 Flashcards - Network Devices

22 flashcards in this section

#	Question / Front	Answer / Back
1	A computer network is a digital telecommunications network which allows [...] to share resources.	nodes
2	[...] and [...] are also known as 'end hosts' or 'endpoints'.	Servers / clients
3	What does this symbol represent? 	Router
4	What does this symbol represent? 	Switch
5	What does this symbol represent? 	Client
6	What does this symbol represent? 	Server
7	What does this symbol represent? 	Firewall
8	A [...] is a device that accesses a service made available by a server.	client
9	A [...] is a device that provides functions or services for clients.	server
10	Can a single device be both a client and a server?	Yes
11	What does LAN stand for?	Local Area Network
12	A [...] provides connectivity to hosts within the same LAN.	switch
13	A [...] has many network interfaces for end hosts to connect to.	switch
14	What kind of network device is a Cisco 'Catalyst'?	Switch
15	What kind of network device is a Cisco ISR?	Router
16	Routers have [...] network interfaces than switches.	fewer
17	A [...] is used to send data over the Internet.	router
18	A [...] monitors and controls network traffic based on configured rules.	firewall
19	[...] firewalls are software applications that filter traffic entering and exiting a host machine, like a PC.	Host-based
20	What kind of network device is a Cisco ASA?	Firewall
21	What kind of network device is a Cisco Firepower?	Firewall
22	A firewall which combines a traditional firewall with more advanced filtering functionalities is known as a '[...] firewall'.	next-generation

Day 02 Flashcards - Interfaces and Cables

57 flashcards in this section

#	Question / Front	Answer / Back
1	What is the name of the connector on the end of an Ethernet UTP cable?	RJ45
2	How many pins are in an RJ45 connector?	8
3	How many bits are in 1 byte?	8
4	Network speeds are measured in [...] per second.	bits
5	How many bits are in 1 kilobit?	1,000 (one thousand)
6	How many bits are in 1 megabit?	1,000,000 (1 million)
7	How many bits are in 1 gigabit?	1,000,000,000 (1 billion)

#	Question / Front	Answer / Back
8	How many bits are in 1 terabit?	1,000,000,000,000 (1 trillion)
9	Ethernet was originally defined in the IEEE [...] standard in 1983.	802.3
10	What is the maximum length of an Ethernet UTP cable?	100 meters
11	What is the speed of an 'Ethernet' connection?	10 Mbps
12	What is the speed of a 'FastEthernet' connection?	100 Mbps
13	What is the speed of a 'Gigabit Ethernet' connection?	1 Gbps
14	What is the speed of a '10 Gig Ethernet' connection?	10 Gbps
15	IEEE standard [...] = 10 Mbps Ethernet (10BASE-T)	802.3i
16	IEEE standard [...] = 100 Mbps Ethernet (100BASE-T)	802.3u
17	IEEE standard [...] = 1 Gbps Ethernet (1000BASE-T)	802.3ab
18	IEEE standard [...] = 10 Gbps Ethernet (10GBASE-T)	802.3an
19	What does UTP stand for?	Unshielded Twisted Pair
20	Why are the wire pairs in a UTP cable twisted together?	It protects against EMI (Electromagnetic Interference)
21	How many pairs of wires are used in a 10BASE-T cable?	2 (=4 wires)
22	How many pairs of wires are used in a 100BASE-T cable?	2 (=4 wires)
23	How many pairs of wires are used in a 1000BASE-T cable?	4 (=8 wires)
24	How many pairs of wires are used in a 10GBASE-T cable?	4 (=8 wires)
25	A [...] connection allows both devices to send and receive data at the same time.	full-duplex
26	In a FastEthernet connection, a router transmits data on which pins [...]	1 and 2
27	In a FastEthernet connection, a router receives data on which pins [...]	3 and 6
28	In a FastEthernet connection, a firewall transmits data on which pins [...]	1 and 2
29	In a FastEthernet connection, a firewall receives data on which pins [...]	3 and 6
30	In a FastEthernet connection, a PC transmits data on which pins [...]	1 and 2
31	In a FastEthernet connection, a PC receives data on which pins [...]	3 and 6
32	In a FastEthernet connection, a switch transmits data on which pins [...]	3 and 6
33	In a FastEthernet connection, a switch receives data on which pins [...]	1 and 2
34	If Auto MDI-X is disabled, what kind of UTP cable should be used to connect a router and a switch?	straight-through cable
35	If Auto MDI-X is disabled, what kind of UTP cable should be used to connect a PC and a switch?	straight-through cable
36	If Auto MDI-X is disabled, what kind of UTP cable should be used to connect two routers together?	crossover cable
37	If Auto MDI-X is disabled, what kind of UTP cable should be used to connect two switches together?	crossover cable
38	What kind of cable connects a pin pair on one end of a UTP cable to the same pair on the other end?	straight-through cable
39	What kind of cable connects a pin pair on one end of a UTP cable to the opposite pair on the other end?	crossover cable
40	What is the name of the feature that allows a device to automatically adjust which RJ45 pin pairs it uses to transmit and receive data?	Auto MDI-X
41	Which two pin pairs are used in a 10BASE-T or 100BASE-T connection?	Pair 1-2, Pair 3-6
42	Which four pin pairs are used in a 1000BASE-T or 10GBASE-T connection?	Pair 1-2, Pair 3-6, Pair 4-5, pair 7-8
43	Fiber-Optic cables are connected to [...] transceivers in a router or switch.	SFP (Small Form-Factor Pluggable)
44	What are the two main kinds of fiber-optic cable?	Single-mode and multimode
45	Multimode Fiber cables have a [...] fiberglass core than single-mode fiber cables.	wider
46	[...] fiber cables allow multiple angles (modes) of light waves to enter the fiberglass core.	Multimode
47	Which allows longer cables, single-mode or multimode fiber?	single-mode
48	Which is cheaper, single-mode or multimode fiber?	multimode
49	In [...] fiber cables, light travels straight through the fiberglass core at a single angle.	single-mode
50	10GBASE-SR, 10GBASE-LR, and 10GBASE-ER were defined in the IEEE [...] standard.	802.3ae
51	1000BASE-LX was defined in the IEEE [...] standard.	802.3z
52	1000BASE-LX maximum cable length:	550 meters (multimode)5 kilometers (single-mode)
53	10GBASE-SR maximum cable length:	400 meters
54	10GBASE-LR maximum cable length:	10 kilometers
55	10GBASE-ER maximum cable length:	30 kilometers
56	[...] cables emit a faint signal outside of the cable, which can be detected and copied.	UTP

#	Question / Front	Answer / Back
57	[...] cables are vulnerable to EMI (electromagnetic interference).	UTP

Day 03 Flashcards - OSI Model, TCP/IP Suite

41 flashcards in this section

#	Question / Front	Answer / Back
1	What does 'OSI' stand for?	Open Systems Interconnection
2	The OSI model is divided into [...] layers.	7
3	Which layer of the OSI model interacts with software applications on a computer?	Layer 7 - Application
4	Which layer of the OSI model includes HTTP and HTTPS?	Layer 7 - Application
5	Which layer of the OSI model identifies communication partners and synchronizes communication?	Application
6	The process of adding additional headers and trailers to data before it is sent over the network is called [...]	encapsulation
7	The process of removing headers and trailers from data received over a network is called [...]	de-encapsulation
8	Interaction between the same OSI layer on different devices is called [...] interaction.	same-layer
9	Interaction between the different OSI layers on the same device is called [...] interaction.	adjacent-layer
10	Which layer of the OSI model translates between data formats?	Layer 6 - Presentation
11	Which layer of the OSI model establishes, manages, and terminates connections between the local and remote applications?	Layer 5 - Session
12	Which layer of the OSI model segments and reassembles data for communications between end hosts?	Layer 4 - Transport
13	Which layer of the OSI model provides host-to-host communication?	Layer 4 - Transport
14	Which layer of the OSI model provides process-to-process communication?	Layer 7 - Application
15	What is the name of the Layer 4 PDU?	Segment
16	Which layer of the OSI model provides path selection between source and destination?	Layer 3 - Network
17	Routers operate at Layer [...] of the OSI model.	3
18	Which layer of the OSI model provides logical addressing (IP addresses)?	Layer 3 - Network
19	Which layer of the OSI model provides connectivity to end hosts on different networks?	Layer 3 - Network
20	What is the name of the Layer 3 PDU?	Packet
21	What is the name of the Layer 2 PDU?	Frame
22	Which layer of the OSI model provides node-to-node connectivity?	Layer 2 - Data Link
23	Which layer of the OSI model detects and corrects Physical layer errors?	Layer 2 - Data Link
24	Switches operate at Layer [...] of the OSI model.	2
25	Which layer of the OSI model defines physical characteristics of the medium used to transfer data between devices?	Layer 1 - Physical
26	What is the name of the Layer 1 PDU?	Bits
27	List the layers of the OSI model in order: 7. [...] 6. [...] 5. [...] 4. Transport 3. Network 2. Data Link 1. Physical	7. Application 6. Presentation 5. Session
28	List the layers of the OSI model in order: 7. Application 6. [...] 5. [...] 4. [...] 3. Network 2. Data Link 1. Physical	6. Presentation 5. Session 4. Transport
29	List the layers of the OSI model in order: 7. Application 6. Presentation 5. [...] 4. [...] 3. [...] 2. Data Link 1. Physical	5. Session 4. Transport 3. Network

#	Question / Front	Answer / Back
30	List the layers of the OSI model in order: 7. Application 6. Presentation 5. Session 4. [...] 3. [...] 2. [...] 1. Physical	4. Transport 3. Network 2. Data Link
31	List the layers of the OSI model in order: 7. Application 6. Presentation 5. Session 4. Transport 3. [...] 2. [...] 1. [...]	3. Network 2. Data Link 1. Physical
32	Which networking model was developed by the United States Department of Defense?	TCP/IP Suite (TCP/IP Model)
33	Which networking model is in use in modern networks?	TCP/IP
34	OSI Model [...] Layer(s) = TCP/IP Application Layer	Application, Presentation, Session
35	OSI Model [...] Layer(s) = TCP/IP Transport Layer	Transport
36	OSI Model [...] Layer(s) = TCP/IP Internet Layer	Network
37	OSI Model [...] Layer(s) = TCP/IP Link Layer	Data Link, Physical
38	OSI Model Application, Presentation, Session Layers = TCP/IP [...] Layer	Application
39	OSI Model Transport Layer = TCP/IP [...] Layer	Transport
40	OSI Model Network Layer = TCP/IP [...] Layer	Internet
41	OSI Model Data Link, Physical Layers = TCP/IP [...] Layer	Link

Day 04 Flashcards - Intro to the CLI

28 flashcards in this section

#	Question / Front	Answer / Back
1	What two kinds of console ports are common on Cisco devices?	USB RJ45
2	What kind of cable is used to connect to the RJ45 console port on a Cisco device?	Rollover cable
3	Default Cisco console port settings: Speed (baud rate) = [...]	9600 bits per second
4	Default Cisco console port settings: Data bits = [...]	8
5	Default Cisco console port settings: Stop bits = [...]	1
6	Default Cisco console port settings: Parity = [...]	
7	Default Cisco console port settings: Flow control = [...]	
8	In Cisco IOS, user EXEC mode is indicated by what? Router[...]	>
9	In Cisco IOS, privileged EXEC mode is indicated by what? Router[...]	#
10	In Cisco IOS, global configuration mode is indicated by what? Router[...]	(config)#
11	What is the lowest-level mode in Cisco IOS?	user EXEC mode
12	In Cisco IOS, what mode provides complete access to view the device's configuration files?	privileged EXEC mode
13	View the available commands: Router# [...]	?
14	Enter privileged EXEC mode: Router> [...]	enable
15	Enter global configuration mode: Router# [...]	configure terminal
16	Configure an unencrypted password to protect privileged EXEC mode: Router(config)# [...]	enable password password
17	Return to privileged EXEC mode: Router(config)# [...]	exit
18	What two configuration files are kept on the device?	running-config, startup-config
19	View the running configuration: Router# [...]	show running-config
20	View the startup configuration: Router# [...]	show startup-config
21	Which three commands can be used to save the configuration? Router# [...] Router# [...] Router# [...]	writewrite memory copy running-config startup-config
22	Encrypt current and future passwords on the device: Router(config)# [...]	service password-encryption
23	Configure an MD5-encrypted enable password: Router(config)# [...]	enable secret password
24	You use the [...] keyword to execute privileged EXEC commands in global configuration mode.	do

#	Question / Front	Answer / Back
25	Use the [...] keyword to remove a configured command.	no (for example, no service password-encryption)
26	Cisco's service password-encryption command uses type [...] encryption.	7
27	enable secret uses type [...] (MD5) encryption.	5
28	If service password-encryption is removed, will currently encrypted passwords be decrypted?	No

Day 05 Flashcards - Ethernet LAN Switching (Part 1)

38 flashcards in this section

#	Question / Front	Answer / Back
1	What is the only field in an Ethernet trailer?	FCS (Frame Check Sequence)
2	What are the 5 fields of an Ethernet header?	Preamble, SFD, Destination, Source, Type/Length
3	What does 'SFD' in an Ethernet header stand for?	Start Frame Delimiter
4	What does 'FCS' in an Ethernet frame stand for?	Frame Check Sequence
5	Ethernet header 'Preamble' = [...] byte(s)	7
6	Ethernet header 'SFD' = [...] byte(s)	1
7	Ethernet header 'Destination' = [...] byte(s)	6
8	Ethernet header 'Source' = [...] byte(s)	6
9	Ethernet header 'Type/Length' = [...] byte(s)	2
10	Ethernet trailer 'FCS' = [...] byte(s)	4
11	Which field of an Ethernet header is used for synchronization?	Preamble
12	What is the bit pattern of one byte in the Ethernet preamble?	10101010
13	What is the bit pattern of the Ethernet SFD?	10101011
14	Which field of an Ethernet header marks the end of the preamble?	SFD
15	What does MAC stand for?	Media Access Control
16	How long is a MAC address? = [...] bytes	6
17	How long is a MAC address? = [...] bits	48
18	A value of [...] or less in the Type/Length field indicates Length.	1500
19	A value of [...] or greater in the Type/Length field indicates Type.	1536
20	What is the Ethernet Type value for IPv4? (in hexadecimal)	0x0800
21	What is the Ethernet Type value for IPv6? (in hexadecimal)	0x86DD
22	What kind of algorithm does the Ethernet FCS use?	CRC (Cyclic Redundancy Check)
23	What is another name for a MAC address?	BIA (Burned-In Address)
24	The first [...] bytes of a MAC address are the OUI.	3
25	The first 3 bytes of a MAC address are the [...]	OUI (Organizationally Unique Identifier)
26	A MAC address is written as [...] hexadecimal characters.	12
27	Hexadecimal A = Decimal [...]	10
28	Hexadecimal B = Decimal [...]	11
29	Hexadecimal C = Decimal [...]	12
30	Hexadecimal D = Decimal [...]	13
31	Hexadecimal E = Decimal [...]	14
32	Hexadecimal F = Decimal [...]	15
33	A [...] frame is a frame destined for a single target.	unicast
34	A MAC address learned by a switch without manual configuration is called a ...	Dynamic MAC address (Dynamically learned MAC address)
35	What does a switch do with an unknown unicast frame?	Floods it out all ports (except the one it was received on)
36	A switch will flood an [...] unicast frame.	unknown
37	By default, a dynamic MAC address will be removed from the MAC address table after [time] of inactivity.	5 minutes
38	Which field of an Ethernet frame does a switch use to learn and populate its MAC address table?	Source MAC Address

Day 06 Flashcards - Ethernet LAN Switching (Part 2)

17 flashcards in this section

#	Question / Front	Answer / Back
1	What is the size of the Ethernet header + trailer? (no Preamble + SFD)	18 bytes
2	What is the minimum size of an Ethernet frame? (Header + Payload + Trailer)	64 bytes
3	What is the minimum size of an Ethernet payload?	46 bytes
4	What protocol is used to discover the Layer 2 address of another device?	ARP (Address Resolution Protocol)
5	What two messages are used in ARP?	ARP RequestARP Reply
6	The ARP Request message is [...]cast.	broad
7	The ARP Reply message is [...]cast.	uni
8	What is the broadcast MAC address?	FFFF.FFFF.FFFF
9	What is the command to view the ARP table on a Windows, macOS, or Linux computer?	arp -a
10	What two messages are used in ping?	ICMP Echo RequestICMP Echo Reply
11	The process of addresses being cleared from the MAC address table after 5 minutes of inactivity is known as [...]	aging
12	Clear all dynamic MAC addresses from the MAC Address Table:SW1# [...]	clear mac address-table dynamic
13	View the MAC Address Table:SW1# [...]	show mac address-table
14	Clear all entries for a specific dynamic MAC address from the MAC Address Table:SW1# [...]	clear mac address-table dynamic address mac-address
15	Clear all dynamic MAC addresses on a specific interface from the MAC Address Table:SW1# [...]	clear mac address-table dynamic interface interface-id
16	What is the Ethernet Type code for an ARP packet?	0x0806
17	Command to show the ARP table on a Cisco device:	show arp

Day 07 Flashcards - IPv4 Addresses (Part 1)

27 flashcards in this section

#	Question / Front	Answer / Back
1	What is the length of an IP address? (bits)	32 bits
2	What is the length of an IP address? (bytes)	4
3	What are the groups of 8 bits in an IP address called?	octets
4	What is the maximum value for a binary octet? (1111 1111)	255
5	What is the numeric range of the first octet of a class A IPv4 address?	0-127
6	What is the numeric range of the first octet of a class B IPv4 address?	128-191
7	What is the numeric range of the first octet of a class C IPv4 address?	192-223
8	What is the numeric range of the first octet of a class D IPv4 address?	224-239
9	What is the numeric range of the first octet of a class E IPv4 address?	240-255
10	What is the binary bit pattern of the first octet of a class A IPv4 address?	0xxxxxxx
11	What is the binary bit pattern of the first octet of a class B IPv4 address?	10xxxxxx
12	What is the binary bit pattern of the first octet of a class C IPv4 address?	110xxxxx
13	What is the binary bit pattern of the first octet of a class D IPv4 address?	1110xxxx
14	What is the binary bit pattern of the first octet of a class E IPv4 address?	1111xxxx
15	What is the class D IPv4 address range reserved for?	Multicast addresses
16	What is the class E IPv4 address range reserved for?	Experimental purposes
17	IPv4 addresses beginning with 127 are used for [...] addresses.	loopback
18	IPv4 addresses beginning with [...] are used for loopback addresses.	127
19	What kind of address is used to test the network software stack on the local device?	loopback address
20	What is the prefix length of a class A IPv4 address?	/8
21	What is the prefix length of a class B IPv4 address?	/16
22	What is the prefix length of a class C IPv4 address?	/24
23	What is the netmask for a /8 prefix length?	255.0.0.0
24	What is the netmask for a /16 prefix length?	255.255.0.0
25	What is the netmask for a /24 prefix length?	255.255.255.0
26	If the host portion of an IPv4 address is all 0s, what kind of address is it?	network address
27	If the host portion of an IPv4 address is all 1s, what kind of address is it?	broadcast address

Day 08 Flashcards - IPv4 Addresses (Part 2)

17 flashcards in this section

#	Question / Front	Answer / Back
1	What is the formula to calculate the maximum number of hosts in a network?	$(2^n) - 2$ (n = number of host bits)
2	Which IPv4 address class contains a total of 128 networks?	Class A
3	Which IPv4 address class contains a total of 16,384 networks?	Class B
4	Which IPv4 address class contains a total of 2,097,152 networks?	Class C
5	Which IPv4 address class allows a total of 16,777,214 hosts per network?	Class A
6	Which IPv4 address class allows a total of 65,534 hosts per network?	Class B
7	Which IPv4 address class allows a total of 254 hosts per network?	Class C
8	If an interface has the shutdown command applied to it, what will the 'status' column of show ip interface brief display?	administratively down
9	The 'Status' field of the show ip interface brief command shows the Layer [...] status of the interface.	1
10	The 'Protocol' field of the show ip interface brief command shows the Layer [...] status of the interface.	2
11	The '['[...]']' field of the show ip interface brief command shows the Layer 1 status of the interface.	Status
12	The '['[...]']' field of the show ip interface brief command shows the Layer 2 status of the interface.	Protocol
13	Interfaces on Cisco [device type] are administratively down by default.	routers
14	Interfaces on Cisco [device type] are NOT administratively down by default.	switches
15	Command to configure an IP address on an interface: R1(config-if)# [...]	ip address ip-address subnet-mask ie. R1(config-if)# ip address 192.168.0.254 255.255.255.0
16	Command to configure a description on an interface: R1(config-if)# [...]	description description ie. R1(config-if)# description ## to SW1 ##
17	Display the configured description of each interface on a router: R1# [...]	show interfaces description

Day 09 Flashcards - Switch Interfaces

23 flashcards in this section

#	Question / Front	Answer / Back
1	In the output of show ip interface brief, router interfaces will be in the [...] state by default.	administratively down/down
2	In the output of show ip interface brief, switch interfaces will be in the [...] state by default if connected to another device.	up/up
3	In the output of show ip interface brief, switch interfaces will be in the [...] state by default if not connected to another device.	down/down
4	What is the default speed setting of an interface?	auto
5	What is the default duplex setting of an interface?	auto
6	Configure a speed of 100 Mbps on an interface: SW1(config-if)# [...]	speed 100
7	Configure full duplex on an interface: SW1(config-if)# [...]	duplex full
8	Which command lets you configure multiple interfaces at the same time?	interface range
9	In [...] duplex mode, an interface can send and receive data at the same time.	full
10	In [...] duplex mode, an interface cannot send and receive data at the same time.	half
11	Ethernet hubs operate at Layer [...] of the OSI model.	1
12	Ethernet devices use [...] to detect and prevent collisions on half-duplex interfaces.	CSMA/CD
13	What does CSMA/CD stand for?	Carrier Sense Multiple Access with Collision Detection
14	All devices connected to an Ethernet hub are in the same [...] domain.	collision
15	Are all devices connected to an Ethernet switch in the same collision domain?	No
16	Devices attached to an Ethernet hub must operate in [...] duplex mode.	half
17	Devices attached to a switch can operate in [...] duplex mode.	full
18	Which show interfaces error counter? -Counts frames that are smaller than the minimum frame size (64 bytes)	runts
19	Which show interfaces error counter? -Counts frames that are larger than the maximum frame size (1518 bytes)	giants
20	Which show interfaces error counter? -Counts frames that failed their error check in the Ethernet FCS trailer.	CRC

#	Question / Front	Answer / Back
21	Which show interfaces error counter? -Counts frames that have an incorrect/illegal format.	frame
22	Which show interfaces error counter? -Total of various counters of errors received by the device	input errors
23	Which show interfaces error counter? -Frames the switch tried to send, but failed due to an error	output errors

Day 10 Flashcards - IPv4 Header

56 flashcards in this section

#	Question / Front	Answer / Back
1	The IPv4 header 'Version' field is [...] bits in length.	4
2	A value of [...] in the IP header 'Version' field indicates IP version 4.	4 (binary 0100)
3	A value of [...] in the IP header 'Version' field indicates IP version 6.	6 (binary 0110)
4	IPv4 header: What does 'IHL' stand for?	Internet Header Length
5	IPv4 header: The 'IHL' field is [...] bits in length.	4
6	IPv4 Header: The IHL field specifies the length of the header in [...] increments.	4-byte (32 bit)
7	IPv4 header: What is the minimum value of the IHL field?	5
8	IPv4 header: What is the maximum value of the IHL field?	15
9	What is the maximum length of the IPv4 header?	60 bytes
10	What is the minimum length of the IPv4 header?	20 bytes
11	IPv4 header: What does 'DSCP' stand for?	Differentiated Services Code Point
12	IPv4 header: The 'DSCP' field is [...] bits in length.	6
13	IPv4 header: Which field is used to prioritize delay-sensitive network traffic?	DSCP
14	IPv4 header: What does 'ECN' stand for?	Explicit Congestion Notification
15	IPv4 header: The 'ECN' field is [...] bits in length.	2
16	IPv4 header: Which field provides end-to-end notification of network congestion?	ECN
17	IPv4 header: The 'Total Length' field is [...] bits in length.	16
18	IPv4 header: Which field indicates the total length of the packet?	Total Length
19	IPv4 header: What is the minimum value of the 'Total Length' field?	20 (=IPv4 header with no encapsulated data)
20	IPv4 header: What is the maximum value of the 'Total Length' field?	65,535
21	IPv4 header: The 'Identification' field is [...] bits in length.	16
22	IPv4 header: Which field is used to identify which packet a fragment belongs to?	Identification
23	IPv4 header: All fragments of the same packet will have the same value in the [...] field.	Identification
24	IPv4 packets are fragmented if they are larger than the [...]	MTU (Maximum Transmission Unit)
25	What does 'MTU' stand for?	Maximum Transmission Unit
26	The MTU is usually [...] bytes.	1500
27	IPv4 header: The 'Flags' field is [...] bits in length.	3
28	Unfragmented IPv4 packets will always have the MF bit set to [...]	0
29	IPv4 header 'Flags' field: Bit [...] = Reserved, always set to 0	0
30	IPv4 header 'Flags' field: Bit [...] = DF bit	1
31	IPv4 header 'Flags' field: Bit [...] = MF bit	2
32	IPv4 header 'Flags' field: What does 'DF bit' stand for?	Don't Fragment bit
33	IPv4 header 'Flags' field: What does 'MF bit' stand for?	More Fragments bit
34	IPv4 header: The 'Fragment Offset' field is [...] bits in length.	13
35	IPv4 header: Which field is used to indicate the position of a fragment within an original, unfragmented IP packet?	Fragment Offset
36	IPv4 header: the 'TTL' field is [...] bits in length.	8
37	The recommended default TTL of an IPv4 packet is [...]	64
38	A router will drop a packet with a TTL of [...]	0
39	IPv4 header: Which field is used to prevent infinite routing loops?	TTL
40	IPv4 header: What does 'TTL' stand for?	Time to Live
41	IPv4 header: The 'Protocol' field is [...] bits in length.	8
42	IPv4 header: Which field indicates the protocol of the encapsulated L4PDU?	Protocol
43	IPv4 header 'Protocol' field: Value of [...] = TCP	6
44	IPv4 header 'Protocol' field: Value of 6 = [...]	TCP
45	IPv4 header 'Protocol' field: Value of [...] = UDP	17

#	Question / Front	Answer / Back
46	IPv4 header 'Protocol' field: Value of 17 = [...]	UDP
47	IPv4 header 'Protocol' field: Value of 1 = [...]	ICMP
48	IPv4 header 'Protocol' field: Value of [...] = ICMP	1
49	IPv4 header 'Protocol' field: Value of 89 = [...]	OSPF
50	IPv4 header 'Protocol' field: Value of [...] = OSPF	89
51	IPv4 header: the 'Checksum' field is [...] bits in length.	16
52	IPv4 header: Which field is used to check for errors in the header?	Header Checksum
53	IPv4 header: the 'Source IP Address' field is [...] bits in length.	32
54	IPv4 header: the 'Destination IP Address' field is [...] bits in length.	32
55	IPv4 header: What is the minimum length of the 'Options' field?	0 bits
56	IPv4 header: What is the maximum length of the 'Options' field?	320 bits / 40 bytes

Day 11 (part 1) Flashcards - Routing Fundamentals

16 flashcards in this section

#	Question / Front	Answer / Back
1	Routers store routes to all of their known destinations in the [...]	routing table
2	When routers receive packets, they look in the [...] to find the best route to forward that packet.	routing table
3	A [...] route is manually configured by a network engineer/admin.	static
4	A [...] route is added to the routing table by a protocol (ie. OSPF) that allows routers to communicate with each other and share routing information.	dynamic
5	The next router in the path to the destination is called the [...].	next hop
6	Show the routing table: R1# [...]	show ip route
7	What does code C in the routing table mean?	Connected
8	What does code L in the routing table mean?	Local
9	Which two routes are automatically added to the routing table when you configure an IP address on an interface?	1) a connected route 2) a local route
10	A [...] route is a route to the actual IP address configured on the interface.	local
11	A [...] route is a route to the network the interface is connected to.	connected
12	If a packet's destination IP is matched by multiple routes, which route will the router select?	The most specific matching route
13	Most specific matching route = the matching route with the [...]	longest prefix length
14	What will a router do if none of its routes match a packet's destination IP address?	Drop the packet
15	Local routes use a prefix length of [...]	/32
16	What prefix length do connected routes use?	The prefix length configured on the interface.

Day 11 (part 2) Flashcards - Static Routing

13 flashcards in this section

#	Question / Front	Answer / Back
1	For an end host like a PC to send packets outside of its local network, it will send the packets to its [...].	default gateway
2	A default route is a route to [network]/[netmask].	0.0.0.0/0
3	The [...] route is the least specific route possible.	default
4	To send a packet to a destination, does a router need a route to every network in the path to the destination?	No (It only needs a route to the destination network)
5	Configure a static route, specifying the next-hop IP: R1(config)# [...]	ip route ip-address netmask next-hop
6	Configure a static route, specifying the exit-interface: R1(config)# [...]	ip route ip-address netmask exit-interface
7	Configure a static route, specifying the exit-interface and next-hop IP: R1(config)# [...]	ip route ip-address netmask exit-interface next-hop
8	Code [...] in the routing table indicates a static route.	S
9	Configure a default route, specifying the next-hop IP: R1(config)# [...]	ip route 0.0.0.0 0.0.0.0 next-hop
10	Static routes in which you specify only the [...] appear as directly-connected in the routing table.	exit-interface
11	A [...] route is often used to direct traffic to the Internet.	default

#	Question / Front	Answer / Back
12	In the routing table, the output "Gateway of last resort is not set" means [...]	a default route has not been configured
13	Code [...] in the routing table indicates a candidate default route.	*

Day 13 Flashcards - Subnetting

21 flashcards in this section

#	Question / Front	Answer / Back
1	A direct, dedicated connection between two routers is called a [...] network.	point-to-point
2	What does CIDR stand for?	Classless Inter-Domain Routing
3	What system replaced the classful IPv4 addressing system?	CIDR
4	When a classful network is split into several smaller networks, the smaller networks are called...	subnetworks / subnets
5	What is the formula to determine the number of USABLE addresses in a network?	$2^n - 2$ = usable addresses (n = number of host bits)
6	What is the dotted decimal equivalent of a /25 prefix length?	255.255.255.128
7	What is the dotted decimal equivalent of a /26 prefix length?	255.255.255.192
8	What is the dotted decimal equivalent of a /27 prefix length?	255.255.255.224
9	What is the dotted decimal equivalent of a /28 prefix length?	255.255.255.240
10	What is the dotted decimal equivalent of a /29 prefix length?	255.255.255.248
11	What is the dotted decimal equivalent of a /30 prefix length?	255.255.255.252
12	What is the dotted decimal equivalent of a /31 prefix length?	255.255.255.254
13	What is the dotted decimal equivalent of a /32 prefix length?	255.255.255.255
14	A subnet mask of 255.255.255.128 is equivalent to / [...]	25
15	A subnet mask of 255.255.255.192 is equivalent to / [...]	26
16	A subnet mask of 255.255.255.224 is equivalent to / [...]	27
17	A subnet mask of 255.255.255.240 is equivalent to / [...]	28
18	A subnet mask of 255.255.255.248 is equivalent to / [...]	29
19	A subnet mask of 255.255.255.252 is equivalent to / [...]	30
20	A subnet mask of 255.255.255.254 is equivalent to / [...]	31
21	A subnet mask of 255.255.255.255 is equivalent to / [...]	32

Day 16 Flashcards - VLANs (Part 1)

12 flashcards in this section

#	Question / Front	Answer / Back
1	A [...] domain is the group of devices which will receive a frame with the destination MAC address FFFF.FFFF.FFFF.	broadcast
2	VLANs are used to split up [...] domains.	broadcast
3	VLANs operate at Layer [...] of the OSI model.	2
4	Will a Layer 2 switch forward traffic between VLANs? (inter-VLAN routing)	no
5	What are the 5 VLANs that exist by default on a Cisco switch?	1, 1002, 1003, 1004, 1005
6	What VLAN are Cisco switch interfaces in by default?	VLAN1
7	An [...] port is a switchport which belongs to a single VLAN, and usually connects to end hosts like PCs.	access
8	Switchports which carry multiple VLANs are called '[...] ports'.	trunk
9	Configure a switch interface to be an access port: SW1(config-if)# [...]	switchport mode access
10	Configure the VLAN of a switch access port: SW1(config-if)# [...]	switchport access vlan vlan-number
11	Create a VLAN: SW1(config)# [...]	vlan vlan-number
12	Configure the name of a VLAN: SW1(config-vlan)# [...]	name name

Day 17 Flashcards - VLANs (Part 2)

33 flashcards in this section

#	Question / Front	Answer / Back
1	VLANs: [...] ports are known as 'tagged' ports.	Trunk
2	VLANs: [...] ports are known as 'untagged' ports.	Access
3	VLANs: Switches will tag frames sent out of [...] ports with a VLAN number.	trunk
4	What is a Cisco proprietary VLAN trunking protocol?	ISL (Inter-Switch Link)

#	Question / Front	Answer / Back
5	What is an industry standard VLAN trunking protocol?	IEEE 802.1Q (dot1q)
6	The 802.1Q tag is inserted after the [...] field of the Ethernet header.	Source MAC Address
7	The 802.1Q tag is [...] bytes in length.	4
8	802.1Q tag: What does TPID stand for?	Tag Protocol Identifier
9	802.1Q tag: What does PCP stand for?	Priority Code Point
10	802.1Q tag: What does DEI stand for?	Drop Eligible Indicator
11	802.1Q tag: What does VID stand for?	VLAN ID
12	802.1Q tag: The TPID field is [...] bits in length.	16
13	802.1Q tag: The TPID field is always set to a value of [...]	0x8100
14	802.1Q tag: The PCP field is [...] bits in length.	3
15	802.1Q tag: The [...] field is use for Class of Service.	PCP
16	802.1Q tag: The DEI field is [...] in length.	1 bit
17	802.1Q tag: The [...] field indicates that a frame can be dropped if the network is congested.	DEI
18	802.1Q tag: The [...] field identifies the VLAN the frame belongs to.	VID
19	VLANs [...] and [...] are reserved and cannot be used.	0 / 4095
20	The range of usable VLANs is [...] to [...]	1 to 4094
21	The Normal VLAN range is [...] to [...]	1 to 1005
22	The Extended VLAN range is [...] to [...]	1006 to 4094
23	The default native VLAN is VLAN [...] on all trunk ports.	1
24	The switch does not add an 802.1Q tag to frames in the [...] VLAN over a trunk link.	native
25	When a switch receives an untagged frame on a trunk port, it assumes the frame belongs to the [...] VLAN.	native
26	Configure the interface as a trunk port:SW1(config-if)# [...]	switchport mode trunk
27	Configure the encapsulation type on a trunk port:SW1(config-if)# [...]	switchport trunk encapsulation encapsulation-type
28	Display all trunk ports on the switch:SW1# [...]	show interfaces trunk
29	Configure the allowed VLANs on a trunk port:SW1(config-if)# [...]	switchport trunk allowed vian allowed-vlans
30	For security purposes, it is best to change the native VLAN to an [...] VLAN.	unused
31	Configure the native VLAN on a trunk port:SW1(config-if)# [...]	switchport trunk native vlan vlan-number
32	Configure the VLAN number on a router subinterface:R1(config-subif)# [...]	encapsulation dot1q vlan-number
33	ROAS involves configuring VLAN tags and IP addresses on router [...].	subinterfaces

Day 18 Flashcards - VLANs (Part 3)

9 flashcards in this section

#	Question / Front	Answer / Back
1	Configure the native VLAN on a router subinterface:R1(config-subif)# [...]	encapsulation dot1q vlan-id native
2	What is another name for a 'Layer 3 switch'?	multilayer switch
3	What is another name for a 'multilayer switch'?	Layer 3
4	An [...] is a virtual interface you can assign an IP address to on a multilayer switch.	SVI
5	What does SVI stand for?	Switch Virtual Interface
6	Reset an interface to its default configuration:R1(config)# [...]	default interface interface
7	Enable Layer 3 routing on a multilayer switch:SW1(config)# [...]	ip routing
8	Configure a switch interface as a routed port:SW1(config-if)# [...]	no switchport
9	SVIs are [shutdown/no shutdown] by default.	shutdown

Day 19 Flashcards - DTP & VTP

44 flashcards in this section

#	Question / Front	Answer / Back
1	[...] is a Cisco proprietary protocol that allows Cisco switches to dynamically determine their interface status (access or trunk) without manual configuration.	DTP
2	What does DTP stand for?	Dynamic Trunking Protocol
3	DTP is [...] by default on all Cisco switch interfaces.	enabled
4	What are the two DTP interface modes?switchport mode dynamic [...] switchport mode dynamic [...]	autodesirable
5	Interfaces on newer Cisco switches default to switchport mode dynamic [...]	auto

#	Question / Front	Answer / Back
6	Interfaces on older Cisco switches default to switchport mode dynamic [...]	desirable
7	A switchport in dynamic [...] mode will actively try to form a trunk with other Cisco switches.	desirable
8	A switchport in dynamic [...] mode will NOT actively try to form a trunk with other Cisco switches.	auto
9	SW1 G0/0 is connected to SW2 G0/0SW1 G0/0 Administrative mode: dynamic desirable SW2 G0/0 Administrative mode: dynamic desirable Operational mode: [...]	trunk
10	SW1 G0/0 is connected to SW2 G0/0SW1 G0/0 Administrative mode: dynamic desirable SW2 G0/0 Administrative mode: trunk Operational mode: [...]	trunk
11	SW1 G0/0 is connected to SW2 G0/0SW1 G0/0 Administrative mode: dynamic desirable SW2 G0/0 Administrative mode: dynamic auto Operational mode: [...]	trunk
12	SW1 G0/0 is connected to SW2 G0/0SW1 G0/0 Administrative mode: dynamic desirable SW2 G0/0 Administrative mode: static access Operational mode: [...]	static access
13	SW1 G0/0 is connected to SW2 G0/0SW1 G0/0 Administrative mode: dynamic auto SW2 G0/0 Administrative mode: trunk Operational mode: [...]	trunk
14	SW1 G0/0 is connected to SW2 G0/0SW1 G0/0 Administrative mode: dynamic auto SW2 G0/0 Administrative mode: dynamic auto Operational mode: [...]	static access
15	SW1 G0/0 is connected to SW2 G0/0SW1 G0/0 Administrative mode: dynamic auto SW2 G0/0 Administrative mode: static access Operational mode: [...]	static access
16	Configure DTP in desirable mode on a switchport:SW1(config-if)# [...]	switchport mode dynamic desirable
17	Configure DTP in auto mode on a switchport:SW1(config-if)# [...]	switchport mode dynamic auto
18	Two commands that disable DTP negotiation on a switchport: SW1(config-if)# switchport mode access OR SW1(config-if)# [...]	switchport nonegotiate
19	Two commands that disable DTP negotiation on a switchport: SW1(config-if)# switchport nonegotiate OR SW1(config-if)# [...]	switchport mode access
20	In DTP trunk encapsulation negotiation, which encapsulation type is favored?	ISL
21	When using ISL, which VLAN are DTP frames sent in?	VLAN1
22	When using 802.1Q, which VLAN are DTP frames sent in?	The native VLAN
23	[...] allows you to configure VLANs on a central server switch, and other switches (clients) will synchronize their VLAN database to the server.	VTP
24	What does VTP stand for?	VLAN Trunking Protocol
25	How many versions of VTP are there?	3
26	What are the three VTP modes?	Server, client, transparent
27	Cisco switches operate in VTP [...] mode by default.	server
28	Cisco switches operate in VTP version [...] by default.	1
29	VTP servers will increase the [...] every time a VLAN is added/modified/deleted.	revision number
30	VTP servers [can/can't] add/modify/delete VLANs.	can
31	VTP clients [can/can't] add/modify/delete VLANs.	can't
32	VTP transparents [can/can't] add/modify/delete VLANs.	can
33	VTP servers [will/won't] synchronize their VLAN database with a VTP server with a higher revision number.	will
34	VTP clients [will/won't] synchronize their VLAN database with a VTP server with a higher revision number.	will
35	VTP transparents [will/won't] synchronize their VLAN database with a VTP server with a higher revision number.	won't
36	Does VTPv1 support the extended VLAN range (1006-4094)?	No
37	Does VTPv2 support the extended VLAN range (1006-4094)?	No
38	Does VTPv3 support the extended VLAN range (1006-4094)?	Yes
39	Does a VTP server maintain its VLAN database in NVRAM?	Yes
40	Does a VTPv1/v2 client maintain its VLAN database in NVRAM?	No
41	Does a VTP transparent maintain its VLAN database in NVRAM?	Yes
42	What are two methods to reset the VTP revision number to 0? [...] OR Change the VTP mode to transparent.	Change the VTP domain to an unused domain.

#	Question / Front	Answer / Back
43	What are two methods to reset the VTP revision number to 0? Change the VTP domain to an unused domain. OR [...]	Change the VTP mode to transparent.
44	Does a VTPv3 client maintain its VLAN database in NVRAM?	Yes

Day 20 Flashcards - STP (Part 1)

36 flashcards in this section

#	Question / Front	Answer / Back
1	If looped Layer 2 broadcast messages accumulate and congest a network, this is called a [...]	broadcast storm
2	When a switch rapidly and continuously updates a MAC address in its MAC address table, alternating between different interfaces, this is known as [...]	MAC address flapping
3	'Classic' Spanning Tree Protocol is IEEE [...]	802.1D
4	Do all switches run STP by default?	Yes
5	The messages sent by STP switches are called:	BPDUs (Bridge Protocol Data Units)
6	What does BPDU stand for?	Bridge Protocol Data Unit
7	How many root bridges are elected in STP?	One
8	Which field in the STP BPDU is used to elect the root bridge?	Bridge ID field
9	STP: The switch with the lowest [...] becomes the root bridge.	Bridge ID
10	STP: All ports on the root bridge are [...] ports (forwarding state)	designated
11	STP: The default bridge priority is [...] (not including Extended System ID)	32768
12	Three fields of the STP Bridge ID: [...] (4 bits) Extended System ID (12 bits) MAC Address (48 bits)	Bridge Priority
13	Three fields of the STP Bridge ID: Bridge Priority (4 bits) [...] (12 bits) MAC Address (48 bits)	Extended System ID (VLAN ID)
14	Three fields of the STP Bridge ID: Bridge Priority (4 bits) Extended System ID (12 bits) [...] (48 bits)	MAC Address
15	Three fields of the STP Bridge ID: Bridge Priority ([...] bits) Extended System ID (12 bits) MAC Address (48 bits)	4
16	Three fields of the STP Bridge ID: Bridge Priority (4 bits) Extended System ID ([...] bits) MAC Address (48 bits)	12
17	Three fields of the STP Bridge ID: Bridge Priority (4 bits) Extended System ID (12 bits) MAC Address ([...] bits)	48
18	Cisco switches use a version of STP called [...], which runs a separate spanning tree instance in each VLAN.	PVST (Per-Vlan Spanning Tree)
19	What does PVST stand for?	Per-VLAN Spanning Tree
20	The STP bridge priority can be changed in units of [...]	4096 Possible values (you don't have to memorize these!): 0, 4096, 8192, 12288, 16384, 20480, 24576, 28672, 32768, 36864, 40960, 45056, 49152, 53248, 57344, or 61440
21	STP Designated ports are in a [...] state.	forwarding
22	STP: When a switch is powered on, it assumes the role of [...] bridge.	root
23	In an 802.1D STP network, which switch(es) send BPDUs?	Root bridge only
24	Interface Speed: 10 Mbps 802.1D STP cost: [...]	100
25	Interface Speed: [...] 802.1D STP cost: 100	10 Mbps
26	Interface Speed: 100 Mbps 802.1D STP cost: [...]	19
27	Interface Speed: [...] 802.1D STP cost: 19	100 Mbps
28	Interface Speed: 1 Gbps 802.1D STP cost: [...]	4
29	Interface Speed: [...] 802.1D STP cost: 4	1 Gbps
30	Interface Speed: 10 Gbps 802.1D STP cost: [...]	2
31	Interface Speed: [...] 802.1D STP cost: 2	10 Gbps

#	Question / Front	Answer / Back
32	STP root port selection:1: Lowest [...] 2: Lowest [...] 3: Lowest [...]	1. root cost 2. neighbor bridge ID 3. neighbor port ID
33	STP: There must be one [...] port per collision domain.	designated
34	STP: There must be one designated port per [...].	collision domain
35	STP Root ports are in a [...] state.	forwarding
36	STP Non-Designated ports are in a [...] state	blocking

Day 21 Flashcards - STP (Part 2)

41 flashcards in this section

#	Question / Front	Answer / Back
1	What are the two 'stable' STP port states?	Blocking, Forwarding
2	What are the two 'transitional' STP port states?	Listening, Learning
3	STP: Non-designated ports are in a [...] state.	Blocking
4	STP: Do interfaces in a blocking state send/receive regular network traffic?	No
5	STP: Do interfaces in a blocking state send BPDUs?	No
6	STP: Do interfaces in a blocking state receive BPDUs?	Yes
7	STP: Do interfaces in a blocking state learn MAC addresses?	No
8	STP default Forward delay timer:	15 seconds
9	STP: Do interfaces in a Listening state send/receive regular network traffic?	No
10	STP: Do interfaces in a Listening state send/receive BPDUs?	Yes
11	STP: Do interfaces in a Listening state learn MAC addresses?	No
12	STP: Do interfaces in a Learning state send/receive regular network traffic?	No
13	STP: Do interfaces in a Learning state send/receive BPDUs?	Yes
14	STP: Do interfaces in a Learning state learn MAC addresses?	Yes
15	STP: Do interfaces in a Forwarding state send/receive regular network traffic?	Yes
16	STP: Do interfaces in a Forwarding state send/receive BPDUs?	Yes
17	STP: Do interfaces in a Forwarding state learn MAC addresses?	Yes
18	STP default hello timer:	2 seconds
19	STP default max age timer:	20 seconds (10*hello timer)
20	STP BPDUs are only forwarded on [...] ports.	designated
21	STP: A forwarding interface [can/can't] move directly to a blocking state.	can
22	STP: A blocking interface [can/can't] move directly to a forwarding state.	can't
23	PVST+ BPDUs are sent to destination MAC address:	0100.0ccc.cccd
24	Standard STP (NOT PVST) BPDUs are sent to destination MAC address:	0180.c200.0000
25	The STP timers on the [...] determine the timers used for all switches in the network.	root bridge
26	[...] allows an STP port to move immediately to the forwarding state.	Portfast
27	If an interface with [...] enabled receives a BPDU from another switch, the interface will be shut down.	BPDU Guard
28	Enable portfast on a switch port:SW1(config-if)# [...]	spanning-tree portfast
29	Enable portfast on all access ports on the switch:SW1(config)# [...]	spanning-tree portfast default
30	SW1(config)# spanning-tree portfast default This command enables portfast on which ports?	All access ports
31	Enable BPDU Guard on a switch port:SW1(config-if)# [...]	spanning-tree bpduguard enable
32	Enable BPDU Guard on all Portfast-enabled interfaces:SW1(config)# [...]	spanning-tree portfast bpduguard default
33	Which STP optional feature prevents a newly connected switch from becoming the root bridge?	Root Guard
34	Which STP optional feature shuts down an interface if it stops receiving BPDUs?	Loop Guard
35	Configure spanning tree in PVST mode:SW1(config)# [...]	spanning-tree mode pvst
36	Configure the spanning-tree root for a VLAN:SW1(config)# [...]	spanning-tree vlan vlan-id root primary
37	spanning-tree vlan vlan-id root primary sets the STP priority to [...], or 4096 less than the current root's priority.	24576
38	Configure the spanning-tree secondary root for a VLAN:SW1(config)# [...]	spanning-tree vlan vlan-id root secondary
39	spanning-tree vlan vlan-id root secondary sets the STP priority to [...]	28672
40	Configure the STP cost of an interface:SW1(config-if)# [...]	spanning-tree vlan vlan-id cost cost
41	Configure the STP priority of an interface:SW1(config-if)# [...]	spanning-tree vlan vlan-id port-priority priority

Day 21-1 Flashcards - PortFast

8 flashcards in this section

#	Question / Front	Answer / Back
1	PortFast allows a port to skip the STP [...] and [...] states.	Listening / Learning
2	A PortFast-enabled port immediately enters the STP [...] state when it comes online.	Forwarding
3	PortFast should not be enabled on a port connected to a [...].	switch
4	Configure PortFast on an individual interface: SW1(config-if)# [...]	spanning-tree portfast [edge]
5	Enable PortFast by default: SW1(config)# [...]	spanning-tree portfast [edge] default
6	Configuring PortFast in global config mode enables it on all [...] ports.	access
7	View detailed STP information about an interface: SW1# [...]	show spanning-tree interface interface-name detail
8	Configure PortFast on a trunk port: SW1(config-if)# [...]	spanning-tree portfast [edge] trunk

Day 21-2 Flashcards - BPDU Guard & BPDU Filter

21 flashcards in this section

#	Question / Front	Answer / Back
1	Does a PortFast-enabled port send BPDUs?	Yes
2	What happens when a PortFast-enabled port receives a BPDU?	It will revert to acting like a regular STP port (without PortFast)
3	[Feature] err-disables a switch port if it receives an STP BPDU.	BPDU Guard
4	If a BPDU Guard-enabled port receives a BPDU, what happens to the port?	It is error-disabled (err-disabled)
5	Enable BPDU Guard on a port: SW1(config-if)# [...]	spanning-tree bpduguard enable
6	Enable BPDU Guard by default: SW1(config)# [...]	spanning-tree portfast [edge] bpduguard default
7	When BPDU Guard is enabled by default, which ports is it active on?	All PortFast-enabled ports
8	Disable BPDU Guard on a specific port (after it has been enabled by default): SW1(config-if)# [...]	spanning-tree bpduguard disable
9	Manually re-enable an err-disabled port: SW1(config-if)# [...] SW1(config-if)# [...]	shutdown no shutdown
10	Which feature can automatically re-enable err-disabled ports?	ErrDisable Recovery
11	View the status of ErrDisable Recovery on the switch: SW1# [...]	show errdisable recovery
12	The default ErrDisable Recovery interval is [time].	300 seconds/5 minutes
13	Modify the ErrDisable Recovery interval: SW1(config)# [...]	errdisable recovery interval seconds
14	Enable ErrDisable Recovery for a particular cause: SW1(config)# [...]	errdisable recovery cause cause (e.g. errdisable recovery cause bpduguard)
15	[Feature] prevents a switch port from sending BPDUs.	BPDU Filter
16	Enable BPDU Filter on a port: SW1(config-if)# [...]	spanning-tree bpdufilter enable
17	Enable BPDU Filter by default: SW1(config)# [...]	spanning-tree portfast [edge] bpdufilter default
18	Disable BPDU Filter on a port (after it has been enabled by default): SW1(config-if)# [...]	spanning-tree bpdufilter disable
19	BPDU Filter has been enabled in interface config mode. What happens if the port receives a BPDU?	The port ignores the BPDU
20	BPDU Filter has been enabled in global config mode. What happens if the port receives a BPDU?	PortFast and BPDU Filter are disabled, and the port operates as a normal STP port
21	When BPDU Filter is enabled by default, which ports is it active on?	All PortFast-enabled ports

Day 21-3 Flashcards - Root Guard

6 flashcards in this section

#	Question / Front	Answer / Back
1	Root Guard disables a port if [...].	it receives a superior BPDU

#	Question / Front	Answer / Back
2	Enable Root Guard on a port: SW1(config-if)# [...]	spanning-tree guard root
3	If a Root Guard-enabled port receives a superior BPDU, it will enter the [...] state.	Broken (Root Inconsistent)
4	In the output of show spanning-tree, what does BKN mean?	Broken (e.g. disabled by Root Guard)
5	In the output of show spanning-tree, what does ROOT_Inc mean?	Root Inconsistent
6	A port disabled by Root Guard will be re-enabled when [...].	it stops receiving superior BPDUs

Day 21-4 Flashcards - Loop Guard

12 flashcards in this section

#	Question / Front	Answer / Back
1	A [...] is a network link where data transmission occurs in only one direction.	unidirectional link
2	Unidirectional links are typically caused by Layer [...] issues.	1
3	[STP optional feature] protects against loops caused by unidirectional links.	Loop Guard
4	When a Loop Guard-enabled port's Max Age counts down to 0, it enters the [...] state.	Broken (Loop Inconsistent)
5	Enable Loop Guard on a port: SW1(config-if)# [...]	spanning-tree guard loop
6	Enable Loop Guard by default: SW1(config)# [...]	spanning-tree loopguard default
7	Disable Loop Guard on a port: SW1(config-if)# [...]	spanning-tree guard none
8	On which ports does the command spanning-tree loopguard default enable Loop Guard?	All ports
9	Loop Guard and Root Guard [can/can't] be enabled on the same port at the same time.	can't
10	If you configure spanning-tree guard loop on a port and then spanning-tree guard root, which feature is active on the port?	Root Guard *the second command replaces the first
11	If you configure spanning-tree loopguard default in global config mode and then spanning-tree guard root on a port, which feature is active on the port?	Root Guard
12	When will a Loop Guard-enabled port automatically recover?	When it starts receiving BPDUs again

Day 22 Flashcards - Rapid STP

39 flashcards in this section

#	Question / Front	Answer / Back
1	Rapid STP is IEEE [...]	802.1w
2	Multiple STP is IEEE [...]	802.1s
3	What is Cisco's upgrade to 802.1D?	PVST+
4	What is Cisco's upgrade to 802.1w?	Rapid PVST+
5	What are the three RSTP port states?	Discarding Learning Forwarding
6	What are the four RSTP port roles?	Root Designated Alternate Backup
7	Which STP optional features were built in to RSTP to allow ports to move directly to a forwarding state (in certain circumstances)?[...] BackboneFast PortFast	UplinkFast
8	Which STP optional features were built in to RSTP to allow ports to move directly to a forwarding state (in certain circumstances)?UplinkFast [...] PortFast	BackboneFast
9	Which STP optional features were built in to RSTP to allow ports to move directly to a forwarding state (in certain circumstances)?UplinkFast BackboneFast [...]	PortFast
10	The RSTP [...] port role is a discarding port that receives a superior BPDU from another switch.	alternate
11	The RSTP [...] port role is a discarding port that receives a superior BPDU from the same switch.	backup
12	The RSTP [...] port role functions as a backup to the root port.	alternate
13	The RSTP [...] port role functions as a backup to a designated port.	backup
14	Enable rapid PVST mode on the switch:SW1(config)# [...]	spanning-tree mode rapid-pvst
15	In RSTP, which switch(es) originate BPDUs?	All switches

#	Question / Front	Answer / Back
16	Is RSTP compatible with classic STP?	Yes (the interface connected to the switch running classic STP will run in classic STP mode)
17	What is the protocol version identifier in a rapid STP BPDU?	2
18	What is the protocol version identifier in a classic STP BPDU?	0
19	In RSTP, a switch considers a neighbor lost if it misses [...] BPDUs.	3
20	What are the three RSTP link types?	EdgePoint-to-point Shared
21	Configure a switchport as an RSTP edge port:SW1(config-if)# [...]	spanning-tree portfast
22	Configure a switchport as an RSTP point-to-point port:SW1(config-if)# [...]	spanning-tree link-type point-to-point
23	Configure a switchport as an RSTP shared port:SW1(config-if)# [...]	spanning-tree link-type shared
24	RSTP [...] link type is used for full-duplex connections.	point-to-point
25	RSTP [...] link type is used for half-duplex connections.	shared
26	RSTP Port Cost10 Mbps = [...]	2,000,000
27	RSTP Port Cost100 Mbps = [...]	200,000
28	RSTP Port Cost1 Gbps = [...]	20,000
29	RSTP Port Cost10 Gbps = [...]	2,000
30	RSTP Port Cost100 Gbps = [...]	200
31	RSTP Port Cost1 Tbps = [...]	20
32	RSTP Port Cost10 Tbps = [...]	2
33	RSTP Port Cost[...] = 2,000,000	10 Mbps
34	RSTP Port Cost[...] = 200,000	100 Mbps
35	RSTP Port Cost[...] = 20,000	1 Gbps
36	RSTP Port Cost[...] = 2,000	10 Gbps
37	RSTP Port Cost[...] = 200	100 Gbps
38	RSTP Port Cost[...] = 20	1 Tbps
39	RSTP Port Cost[...] = 2	10 Tbps

Day 23 Flashcards - EtherChannel

18 flashcards in this section

#	Question / Front	Answer / Back
1	What does LAG (EtherChannel) stand for?	Link Aggregation Group
2	What are the 6 EtherChannel load-balancing methods? (not including TCP/IP port numbers)	Source MACDestination MAC Source/Destination MAC Source IP Destination IP Source/Destination IP
3	View the current EtherChannel load-balancing method on the switch:SW1# [...]	show etherchannel load-balance
4	Configure the EtherChannel load-balancing method on the switch:SW1(config)# [...]	port-channel load-balance method
5	What does PAgP stand for?	Port Aggregation Protocol
6	What does LACP stand for?	Link Aggregation Control Protocol
7	Which method of EtherChannel configuration uses a Cisco proprietary protocol?	PAgP
8	Which method of EtherChannel configuration uses an industry standard protocol?	LACP
9	LACP is IEEE [...]	802.3ad
10	How many interfaces can be active in an EtherChannel?	8
11	Which EtherChannel protocol allows up to 8 active interfaces and up to 8 standby interfaces?	LACP
12	The EtherChannel channel-group number [does/doesn't] have to match the number on the neighbor switch.	doesn't
13	Which two EtherChannel modes does PAgP use?	Desirable, Auto
14	Which two EtherChannel modes does LACP use?	Active, Passive
15	Configure an interface to join an EtherChannelSW1(config-if)# [...]	channel-group number mode mode
16	Configure static EtherChannel:SW1(config-if)# channel-group 1 mode [...]	on
17	Manually configure the EtherChannel protocol used on an interface:SW1(config-if)# [...]	channel-protocol [lacp pagp]
18	Configure a Layer 3 EtherChannel:SW1(config-if)# [...] SW1(config-if)# channel-group group mode mode	no switchport

Day 24 Flashcards - Dynamic Routing

51 flashcards in this section

#	Question / Front	Answer / Back
1	A route to a /32 destination is known as a [...] route.	host
2	A route to a network/subnet is known as a [...] route.	network
3	Routers can use [...] to advertise information about the routes they know to other routers.	dynamic routing protocols
4	If R1 learns multiple routes to the same destination from two different routing protocols, what is used to determine which route will be added to R1's routing table?	AD (Administrative Distance)
5	If R1 learns multiple routes to the same destination from the same routing protocol, what is used to determine which route will be added to R1's routing table?	Metric
6	Does a router prefer a lower or higher AD when selecting routes?	lower
7	Does a router prefer a lower or higher metric when selecting routes?	lower
8	Dynamic routing: What does IGP stand for?	Interior Gateway Protocol
9	Dynamic routing: What does EGP stand for?	Exterior Gateway Protocol
10	IGPs are used to share routes within a single [...]	autonomous system (AS)
11	[...] are used to share routes within a single autonomous system (AS).	IGPs
12	EGPs are used to share routes between different [...]	autonomous systems (AS)
13	[...] are used to share routes between different autonomous systems (AS).	EGPs
14	What are the two types of IGP?	Distance VectorLink State
15	What is the single type of EGP?	Path Vector
16	What are the two Distance Vector routing protocols?	RIPEIGRP
17	What are the two Link State routing protocols?	OSPFIS-IS
18	What is the single Path Vector routing protocol?	BGP
19	What does RIP stand for?	Routing Information Protocol
20	What does EIGRP stand for?	Enhanced Interior Gateway Routing Protocol
21	What does OSPF stand for?	Open Shortest Path First
22	What does IS-IS stand for?	Intermediate System to Intermediate System
23	What does BGP stand for?	Border Gateway Protocol
24	What kind of dynamic routing protocol operates using 'routing by rumor'?	Distance Vector
25	[...] routing protocols send the following information to their connected neighbors:-their known destination networks -their metric to reach their known destination networks	Distance Vector
26	When using a [...] routing protocol, every router creates a 'connectivity map' of the network.	link state
27	Which kind of routing protocol uses more CPU resources on the router, distance vector or link state?	link state
28	Which kind of dynamic routing protocol tends to react faster to changes in the network, distance vector or link state?	link state
29	What is the metric value of this route?O 192.168.4.0/24 [110/3] via 10.0.12.2	3
30	What is the AD value of this route?O 192.168.4.0/24 [110/3] via 10.0.12.2	110
31	What does ECMP stand for?	Equal Cost Multi-Path (load-balancing)
32	R1 learns two routes to the same destination, via the same routing protocol, and both routes have the same metric. Which will be added to the routing table?	Both (traffic will be load-balanced)
33	What metric does RIP use?	hop count
34	What dynamic routing protocol calculates metric using bandwidth and delay (and others, if configured)?	EIGRP
35	Which dynamic routing protocol uses a cost calculated based on interface bandwidth as its metric?	OSPF
36	In which dynamic routing protocol do all links have a metric cost of 10 by default?	IS-IS
37	[...] is used to compare routes learned via the same routing protocol.	Metric
38	[...] is used to compare routes learned via different routing protocols.	AD (Administrative Distance)
39	Route protocol/type: Directly connectedAD: [...]	0
40	Route protocol/type: StaticAD: [...]	1
41	Route protocol/type: eBGPAD: [...]	20
42	Route protocol/type: EIGRPAD: [...]	90
43	Route protocol/type: IGRPAD: [...]	100
44	Route protocol/type: OSPFAD: [...]	110
45	Route protocol/type: IS-ISAD: [...]	115
46	Route protocol/type: RIPAD: [...]	120

#	Question / Front	Answer / Back
47	Route protocol/type: EIGRP (external)AD: [...]	170
48	Route protocol/type: iBGPAD: [...]	200
49	Route protocol/type: Unusable routeAD: [...]	255
50	Configure a static route, and specify the AD:R1(config)# [...]	ip route destination mask next-hop AD (ie. ip route 10.0.0.0 255.0.0.0 10.0.13.2 115)
51	What is the name for a static route which has a higher AD than a route learned via a dynamic routing protocol?	Floating static route

Day 25 Flashcards (1) - RIP & EIGRP

31 flashcards in this section

#	Question / Front	Answer / Back
1	The maximum hop count for a RIP route is [...]	15
2	Which versions of RIP are used for IPv4?	RIPv1 and RIPv2
3	Which version of RIP is used for IPv6?	RIPng (RIP Next Generation)
4	RIP uses two message types:-[...] -[...]	RequestResponse
5	RIPv[...] only advertises classful addresses.	1
6	RIPv[...] doesn't support VLSM and CIDR.	1
7	RIPv[...] supports VLSM and CIDR.	2
8	RIPv1 messages are sent using [...]cast.	broad
9	RIPv2 messages are sent using [...]cast.	multi
10	RIPv2 messages are multicast to [IPv4 address].	224.0.0.9
11	Enter RIP configuration mode:R1(config)# [...]	router rip
12	Enable RIPv2:R1(config-router)# [...]	version 2
13	Disable RIP auto summarizationR1(config-router)# [...]	no auto-summary
14	Enable RIP on all interfaces in the specified range:R1(config-router)# [...]	network ip-address
15	Configure the router to stop sending RIP advertisements out of a RIP-enabled interface: R1(config-router)# [...]	passive-interface interface-id
16	Configure a RIP-enabled router to advertise its default route to its RIP neighbors:R1(config-router)# [...]	default-information originate
17	EIGRP messages are multicast to [IPv4 address]	224.0.0.10
18	RIP [can/can't] perform unequal-cost load-balancing.	can't
19	EIGRP [can/can't] perform unequal-cost load-balancing.	can
20	OSPF [can/can't] perform unequal-cost load-balancing.	can't
21	Modify RIP's AD on the local router:R1(config-router)# [...]	distance value
22	Modify the number of paths RIP will perform ECMP load-balancing over:R1(config-router)# [...]	maximum-paths number
23	Enter EIGRP configuration mode:R1(config)# [...]	router eigrp as-number
24	Disable EIGRP auto summarization:R1(config-router)# [...]	no auto-summary
25	Does the EIGRP AS number have to match between EIGRP neighbors?	Yes
26	Prevent the router from sending EIGRP messages out of an EIGRP-enabled interface: R1(config-router)# [...]	passive-interface interface-id
27	Enable EIGRP on interfaces in the specified range: R1(config-router)# [...]	network ip-address [wildcard-mask]
28	EIGRP Router ID selection priority:1. [...] 2. Highest IP address on a loopback interface 3. Highest IP address on a physical interface	Manual configuration
29	EIGRP Router ID selection priority:1. Manual configuration 2. [...] 3. Highest IP address on a physical interface	Highest IP address on a loopback interface
30	EIGRP Router ID selection priority:1. Manual configuration 2. Highest IP address on a loopback interface 3. [...]	Highest IP address on a physical interface
31	Configure the EIGRP router ID:R1(config-router)# [...]	eigrp router-id a.b.c.d

Day 25 Flashcards (2 - Lab Video) - EIGRP Terms

12 flashcards in this section

#	Question / Front	Answer / Back
1	Which EIGRP term?[...] = This router's metric value to the route's destination	Feasible Distance
2	Which EIGRP term?[...] = The neighbor's metric value to the route's destination.	Reported Distance (aka Advertised Distance)
3	Which EIGRP term?[...] = The route with the lowest metric to the destination (the best route)	Successor
4	Which EIGRP term?[...] = An alternate route to the destination which meets the feasibility condition	Feasible Successor
5	EIGRP feasibility condition:A route is considered a [...] if its reported distance is lower than the successor route's feasible distance.	feasible successor
6	EIGRP feasibility condition:A route is considered a feasible successor if its [...] is lower than the successor route's feasible distance.	reported distance
7	EIGRP feasibility condition:A route is considered a feasible successor if its reported distance is lower than the [...] route's feasible distance.	successor
8	EIGRP feasibility condition:A route is considered a feasible successor if its reported distance is lower than the successor route's [...].	feasible distance
9	EIGRP will only perform unequal-cost load-balancing over [...] routes.	feasible successor
10	If an EIGRP route doesn't meet the feasibility requirement, can it be used for unequal-cost load-balancing?	No
11	EIGRP default K-values:K1 = [...] K2 = [...] K3 = [...] K4 = [...] K5 = [...]	K1 = 1 K2 = 0 K3 = 1 K4 = 0 K5 = 0
12	EIGRP metric calculation uses the bandwidth of [...] plus the delay of [...].	the slowest link / all links

Day 26 Flashcards - OSPF (Part 1)

34 flashcards in this section

#	Question / Front	Answer / Back
1	The Shortest Path First algorithm is also known as [...]s algorithm.	Dijkstra
2	Which version of OSPF is usually used for IPv4?	OSPFv2
3	Which version of OSPF is used for IPv6?	OSPFv3
4	OSPF: What does LSA stand for?	Link State Advertisement
5	OSPF: What does LSDB stand for?	Link State Database
6	OSPF [...] are organized in a structure called the LSDB.	LSAs
7	OSPF LSAs are organized in a structure called the [...].	LSDB
8	OSPF LSAs have a default aging timer of [time]	30 minutes
9	OSPF can divide networks into separate [...]	areas
10	OSPF: An [...] is a set of routers and links that share the same LSDB.	area
11	OSPF: The [...] area (area 0) is an area that all other areas must connect to.	backbone
12	OSPF: The backbone area (area [...]) is an area that all other areas must connect to.	0
13	OSPF: Routers with all interfaces in the same area are called [...] routers.	internal
14	OSPF: Routers with interfaces in multiple areas are called [...].	area border routers (ABRs)
15	OSPF: What does ABR stand for?	Area Border Router
16	OSPF: It is recommend that you connect an ABR to a maximum of [...] areas.	2
17	OSPF: Routers connected to the backbone area (area 0) are called [...] routers.	backbone
18	OSPF: An [...] route is a route to a destination inside the same OSPF area.	intra-area
19	OSPF: An [...] route is a route to a destination in a different OSPF area.	interarea
20	Enter OSPF configuration mode:R1(config)# [...]	router ospf process-id
21	Enable OSPF on interfaces in the specified range: R1(config-router)# [...]	network ip-address wildcard-mask area area
22	Prevent an OSPF router from sending OSPF messages out of an OSPF-enabled interface: R1(config-router)# [...]	passive-interface interface-id
23	Configure an OSPF router to advertise its default route into the OSPF domain: R1(config-router)# [...]	default-information originate
24	Modify the OSPF AD on the local router: R1(config-router)# [...]	distance value
25	Configure the maximum number of paths an OSPF router will use to perform ECMP load-balancing: R1(config-router)# [...]	maximum-paths number

#	Question / Front	Answer / Back
26	OSPF Router ID selection priority: 1. [...] 2. Highest IP address on a loopback interface 3. Highest IP address on a physical interface	Manual configuration
27	OSPF Router ID selection priority: 1. Manual configuration 2. [...] 3. Highest IP address on a physical interface	Highest IP address on a loopback interface
28	OSPF Router ID selection priority: 1. Manual configuration 2. Highest IP address on a loopback interface 3. [...]	Highest IP address on a physical interface
29	Reset the OSPF process on the local router:R1# [...]	clear ip ospf process
30	OSPF: What does ASBR stand for?	Autonomous System Boundary Router
31	An [...] is an OSPF router that connects the OSPF network to an external network.	autonomous system boundary router (ASBR)
32	Does single-area OSPF have to use area 0?	No
33	Does the OSPF process ID have to match for two routers to become OSPF neighbors?	No
34	Manually configure the OSPF router ID:R1(config-router)# [...]	router-id a.b.c.d

Day 27 Flashcards - OSPF (Part 2)

47 flashcards in this section

#	Question / Front	Answer / Back
1	What is the formula to calculate OSPF interface cost?	Reference bandwidth / interface bandwidth = cost
2	What is the default OSPF reference bandwidth?	100 mbps
3	What is the OSPF cost of an Ethernet (10 mbps) interface with the default reference bandwidth?	10
4	What is the OSPF cost of a FastEthernet (100 mbps) interface with the default reference bandwidth?	1
5	What is the OSPF cost of a Gigabit Ethernet interface with the default reference bandwidth?	1
6	What is the OSPF cost of a 10Gig Ethernet interface with the default reference bandwidth?	1
7	Configure the OSPF reference bandwidth:R1(config-router)# [...]	auto-cost reference-bandwidth megabits-per-second
8	What is the OSPF cost of a loopback interface?	1
9	Configure the OSPF cost of an interface:R1(config-if)# [...]	ip ospf cost cost
10	Configure the bandwidth of an interface:R1(config-if)# [...]	bandwidth kilobits-per-second
11	Does the bandwidth command change the speed at which an interface operates?	No(the speed command does)
12	What is the default OSPF Hello timer on an Ethernet connection?	10 seconds
13	What is the default OSPF Dead timer on an Ethernet connection?	40 seconds
14	What is the multicast address for all OSPF routers?	224.0.0.5
15	OSPF Hello messages are multicast to [...]	224.0.0.5
16	OSPF neighbor states:1. [...] 2. [...] 3. [...] 4. Exstart 5. Exchange 6. Loading 7. Full	DownInit 2-way
17	OSPF neighbor states:1. Down 2. [...] 3. [...] 4. [...] 5. Exchange 6. Loading 7. Full	Init2-way Exstart
18	OSPF neighbor states:1. Down 2. Init 3. [...] 4. [...] 5. [...] 6. Loading 7. Full	2-wayExstart Exchange

#	Question / Front	Answer / Back
19	OSPF neighbor states:1. Down 2. Init 3. 2-way 4. [...] 5. [...] 6. [...] 7. Full	Exstart Exchange Loading
20	OSPF neighbor states:1. Down 2. Init 3. 2-way 4. Exstart 5. [...] 6. [...] 7. [...]	Exchange Loading Full
21	If no Hello messages have been received, the OSPF neighbor state is [...].	Down
22	If a router receives an OSPF Hello packet but its own router ID is not in the Hello packet, the neighbor state is [...].	Init
23	When an OSPF router receives a Hello packet with its own router ID, the OSPF neighbor state is [...].	2-way
24	OSPF DR/BDRs are elected in the [...] neighbor state.	2-way
25	In which OSPF neighbor state are a Master and Slave determined?	Exstart
26	In the OSPF Exstart state, which router becomes the Master?	The router with the higher router ID.
27	In the OSPF Exstart state, which router becomes the Slave?	The router with the lower router ID
28	In the OSPF [...] state, routers exchange DBDs which contain a list of LSAs in their LSDB.	Exchange
29	OSPF: What does DBD stand for?	Database Description
30	OSPF: What does LSR stand for?	Link State Request
31	OSPF: What does LSU stand for?	Link State Update
32	OSPF: What does LSAck stand for?	Link State Acknowledgement
33	In the OSPF [...] state, routers send LSR messages to request LSAs from their neighbor.	Loading
34	OSPF LSAs are sent in [...] messages.	LSU
35	In the OSPF [...] state, the routers have a full OSPF adjacency and identical LSDBs.	Full
36	OSPF Message type [...] = Hello	1
37	OSPF Message type 1 = [...]	Hello
38	OSPF Message type [...] = Database Description (DBD)	2
39	OSPF Message type 2 = [...]	Database Description (DBD)
40	OSPF Message type [...] = Link-State Request (LSR)	3
41	OSPF Message type 3 = [...]	Link-State Request (LSR)
42	OSPF Message type [...] = Link-State Update (LSU)	4
43	OSPF Message type 4 = [...]	Link-State Update (LSU)
44	OSPF Message type [...] = Link-State Acknowledgement (LSAck)	5
45	OSPF Message type 5 = [...]	Link-State Acknowledgement (LSAck)
46	Enable OSPF directly on an interface:R1(config-if)# [...]	ip ospf process-id area area-id
47	Configure all interfaces to be OSPF passive interfaces by default:R1(config-router)# [...]	passive-interface default

Day 28 Flashcards - OSPF (Part 3)

40 flashcards in this section

#	Question / Front	Answer / Back
1	OSPF: The [...] network type is enabled on Ethernet and FDDI interfaces by default.	Broadcast
2	OSPF: The Broadcast network type is enabled on [...] and [...] interfaces by default.	Ethernet / FDDI
3	OSPF: The [...] network type is enabled on PPP and HDLC interfaces by default.	Point-to-point
4	OSPF: The Point-to-point network type is enabled on [...] and [...] interfaces by default.	PPP / HDLC
5	OSPF: The [...] network type is enabled on Frame Relay and X.25 interfaces by default.	Non-broadcast
6	OSPF: The Non-broadcast network type is enabled on [...] and [...] interfaces by default.	Frame Relay / X.25
7	In the OSPF Broadcast network type, do routers dynamically discover neighbors?	Yes
8	In the OSPF Point-to-point network type, do routers dynamically discover neighbors?	Yes
9	In the OSPF Non-broadcast network type, do routers dynamically discover neighbors?	No
10	In the OSPF Broadcast network type, are a DR and BDR elected?	Yes
11	In the OSPF Point-to-point network type, are a DR and BDR elected?	No
12	OSPF: In the Broadcast network type, routers which aren't the DR/BDR become a [...].	DROther

#	Question / Front	Answer / Back
13	OSPF DR/BDR election order of priority:1: [...] 2: Highest OSPF router ID	Highest OSPF interface priority
14	OSPF DR/BDR election order of priority:1: Highest OSPF interface priority 2: [...]	Highest OSPF router ID
15	The default OSPF interface priority is [...] on all interfaces.	1
16	OSPF DROthers will only move to the Full state with [...].	the DR and BDR
17	OSPF DROthers will remain in the [...] neighbor state with other DROthers.	2-way
18	Messages to the OSPF DR/BDR are multicast using address [...].	224.0.0.6
19	Serial interfaces use the [...] encapsulation by default.	HDLC (cHDLC)
20	Configure the operating speed of a serial interface:R1(config-if)# [...]	clock rate bits-per-second
21	Serial connection: What does DCE stand for?	Data Communications Equipment
22	Serial connection: What does DTE stand for?	Data Terminal Equipment
23	Serial connection: The [...] side of the connection specifies the clock rate.	DCE
24	Configure the PPP encapsulation on a serial interface:R1(config-if)# [...]	encapsulation ppp
25	Which command can be used to check if a serial interface is DCE or DTE?	show controllers interface-id
26	Configure the OSPF network type of an interface:R1(config-if)# [...]	ip ospf network type
27	OSPF Non-broadcast network type default timers:Hello: [...] Dead: [...]	30 seconds120 seconds
28	Disable the OSPF process on the router:R1(config-router)# [...]	shutdown
29	Can routers with matching OSPF router IDs become neighbors?	No
30	Do OSPF Hello and Dead timers have to match for routers to become OSPF neighbors?	Yes
31	Enable OSPF authentication on an interface:R1(config-if)# [...]	ip ospf authentication
32	Configure an OSPF password on an interface:R1(config-if)# [...]	ip ospf authentication-key password
33	If the IP [...] settings on the interfaces don't match, routers can become OSPF neighbors but won't move to the Full state.	MTU
34	If two routers have an OSPF network type mismatch, they will form an OSPF adjacency but won't [...].	learn OSPF routes
35	OSPF Type [...] LSA = Router LSA	1
36	OSPF Type 1 LSA = [...] LSA	Router
37	OSPF Type [...] LSA = Network LSA	2
38	OSPF Type 2 LSA = [...] LSA	Network
39	OSPF Type [...] LSA = AS-External LSA	5
40	OSPF Type 5 LSA = [...] LSA	AS-External

Day 29 Flashcards - FHRPs

36 flashcards in this section

#	Question / Front	Answer / Back
1	What does 'FHRP' stand for?	First Hop Redundancy Protocol
2	What does 'HSRP' stand for?	Hot Standby Router Protocol
3	What does 'VRRP' stand for?	Virtual Router Redundancy Protocol
4	What does 'GLBP' stand for?	Gateway Load Balancing Protocol
5	FHRP-activated routers communicate with each other by sending [...]cast Hello messages.	multi
6	FHRP-activated routers communicate with each other by sending multicast [...] messages.	Hello
7	In [...] ARP, an ARP reply is sent without being requested.	Gratuitous
8	Gratuitous ARP messages are [...]cast.	broad
9	When the HSRP standby router switches to the active role, it will send [...] ARP messages.	gratuitous
10	When using an FHRP, what IP address should be configured as the default gateway for hosts in the subnet?	The virtual IP address
11	When using an FHRP, what MAC address does the active router use to respond to ARP requests?	The virtual MAC address
12	Is HSRP Cisco proprietary?	Yes
13	Is VRRP Cisco proprietary?	No
14	Is GLBP Cisco proprietary?	Yes
15	HSRP uses [...] and [...] routers.	Active / standby
16	VRRP uses [...] and [...] routers.	Master / Backup
17	HSRPv1 multicast IPv4 address:	224.0.0.2
18	HSRPv2 multicast IPv4 address:	224.0.0.102

#	Question / Front	Answer / Back
19	VRRP multicast IPv4 address:	224.0.0.18
20	GLBP multicast IPv4 address:	224.0.0.102
21	HSRPv1 virtual MAC address format:	0000.0c07.acXX(XX = HSRP group number)
22	HSRPv2 virtual MAC address format:	0000.0c9f.fXXX(XXX = HSRP group number)
23	VRRP virtual MAC address format:	0000.5e00.01XX(XX = VRRP group number)
24	GLBP virtual MAC address format:	0007.b400.XXYY(XX = GLBP group number, YY = AVF number)
25	Which FHRP load balances among multiple routers within a single subnet?	GLBP
26	GLBP: A single [...] is elected.	Active Virtual Gateway (AVG)
27	GLBP: The AVG assigns up to [...] Active Virtual Forwarders (AVFs).	four
28	GLBP: The AVG assigns up to four [...].	Active Virtual Forwarders (AVFs)
29	Is preemption enabled on FHRPs by default?	No
30	The HSRP active router is determined in this order: 1 – [...] 2 – Highest IP address	Highest priority
31	The HSRP active router is determined in this order: 1 – Highest priority 2 – [...]	Highest IP address
32	What is the default HSRP priority?	100
33	Configure the HSRP virtual IP on an interface:R1(config-if)# [...]	standby group-number ip ip-address
34	Configure the HSRP priority:R1(config-if)# [...]	standby group-number priority priority
35	Enable HSRP preemption:R1(config-if)# [...]	standby group-number preempt
36	Configure HSRP version 2:R1(config-if)# [...]	standby version 2

Day 30 Flashcards - TCP & UDP

67 flashcards in this section

#	Question / Front	Answer / Back
1	FTP data = TCP port [...]	20
2	[...] = TCP port 20	FTP data
3	FTP data = [...] port 20	TCP
4	[...] = TCP port 21	FTP control
5	FTP control = [...] port 21	TCP
6	FTP control = TCP port [...]	21
7	[...] = TCP port 22	SSH
8	SSH = [...] port 22	TCP
9	SSH = TCP port [...]	22
10	[...] = TCP port 23	Telnet
11	Telnet = TCP port [...]	23
12	[...] = TCP port 25	SMTP
13	SMTP = [...] port 25	TCP
14	SMTP = TCP port [...]	25
15	[...] = TCP port 80	HTTP
16	HTTP = [...] port 80	TCP
17	HTTP = TCP port [...]	80
18	[...] = TCP port 110	POP3
19	POP3 = [...] port 110	TCP
20	POP3 = TCP port [...]	110
21	[...] = TCP port 443	HTTPS
22	HTTPS = [...] port 443	TCP
23	HTTPS = TCP port [...]	443
24	[...] = UDP port 67	DHCP server
25	DHCP server = [...] port 67	UDP
26	DHCP server = UDP port [...]	67
27	[...] = UDP port 68	DHCP client
28	DHCP client = UDP port [...]	68
29	DHCP client = [...] port 68	UDP

#	Question / Front	Answer / Back
30	[...] = UDP port 69	TFTP
31	TFTP = [...] port 69	UDP
32	TFTP = UDP port [...]	69
33	[...] = UDP port 161	SNMP agent
34	SNMP agent = [...] port 161	UDP
35	SNMP agent = UDP port [...]	161
36	[...] = UDP port 162	SNMP manager
37	SNMP manager = [...] port 162	UDP
38	SNMP manager = UDP port [...]	162
39	[...] = UDP port 514	Syslog
40	Syslog = [...] port 514	UDP
41	Syslog = UDP port [...]	514
42	[...] = TCP/UDP port 53	DNS
43	DNS = [...] port 53	TCP/UDP
44	DNS = TCP/UDP port [...]	53
45	Telnet = [...] port 23	TCP
46	What does TCP stand for?	Transmission Control Protocol
47	What does UDP stand for?	User Datagram Protocol
48	What are Layer 4 addresses?	Port numbers
49	[TCP/UDP] provides reliable data transfer.	TCP
50	[TCP/UDP] provides error recovery.	TCP
51	[TCP/UDP] provides data sequencing.	TCP
52	[TCP/UDP] provides flow control.	TCP
53	[TCP/UDP] has less overhead.	UDP
54	The Layer 4 destination port number indicates the [...] Layer protocol.	Application
55	The Layer 4 [source/destination] port number is randomly selected by the host that initiates the data exchange.	source
56	Layer 4 well-known port range (according to IANA):	0 - 1023
57	Layer 4 registered port range (according to IANA):	1024 - 49151
58	Layer 4 ephemeral port range (according to IANA):	49152 - 65535
59	The TCP header [...] field is used for flow control.	window size
60	TCP three-way handshake order of messages (establishing a connection):	SYN ->- SYN-ACK ACK ->
61	TCP four-way handshake order of messages (terminating a connection):	FIN ->- ACK -<- FIN ACK ->
62	[TCP/UDP] is connection-oriented.	TCP
63	[TCP/UDP] is connectionless.	UDP
64	TCP: [...] is used to indicate the sequence number of the next segment the host expects to receive.	Forward acknowledgement
65	TCP: A '[' [...] ']' is used to dynamically adjust how large the window size is.	sliding window
66	For applications like real-time voice and video, [TCP/UDP] is preferred.	UDP
67	For applications that require reliable communications, [TCP/UDP] is preferred.	TCP

Day 31 Flashcards - IPv6 (Part 1)

13 flashcards in this section

#	Question / Front	Answer / Back
1	Which RIR controls IP address assignments in Africa?	AFRINIC
2	Which RIR controls IP address assignments in Asia-Pacific?	APNIC
3	Which RIR controls IP address assignments in Canada, many Caribbean and North Atlantic islands, and the US?	ARIN
4	Which RIR controls IP address assignments in Latin America and the Caribbean?	LACNIC
5	Which RIR controls IP address assignments in Europe, the Middle East, and parts of Central Asia?	RIPE NCC
6	An IPv6 address is [...] bits in length.	128
7	An IPv6 address contains [...] hexadecimal quartets.	8
8	To shorten an IPv6 address, you can remove [...] from each quartet.	leading 0s

#	Question / Front	Answer / Back
9	To shorten an IPv6 address, you can replace consecutive quartets of all 0s with a [...].	double colon (::)*can only do this once per address
10	Typically, an enterprise requesting IPv6 addresses will receive a [prefix length] block of addresses.	/48
11	Typically, IPv6 subnets use a [...] prefix length.	/64
12	Enable IPv6 routing on the router.R1(config)# [...]	ipv6 unicast-routing
13	Configure an IPv6 address on an interfaceR1(config-if)# [...]	ipv6 address address/prefix-length

Day 32 Flashcards - IPv6 (Part 2)

75 flashcards in this section

#	Question / Front	Answer / Back
1	What does EUI stand for?	Extended Unique Identifier
2	3 steps to convert a MAC address into an EUI-64 identifier:1. [...] 2. Insert FFFE in the middle 3. Invert the 7th bit	Divide the MAC address in half
3	3 steps to convert a MAC address into an EUI-64 identifier:1. Divide the MAC address in half 2. [...] 3. Invert the 7th bit	Insert FFFE in the middle
4	3 steps to convert a MAC address into an EUI-64 identifier:1. Divide the MAC address in half 2. Insert FFFE in the middle 3. [...]	Invert the 7th bit
5	Configure an IPv6 address using EUI-64R1(config-if)# [...]	ipv6 address prefix/prefix-length eui-64
6	MAC addresses: What does UAA stand for?	Universally Administered Address
7	MAC addresses: What does LAA stand for?	Locally Administered Address
8	What is the 7th bit of a MAC address called?	U/L bit (Universal/Local bit)
9	If the U/L bit of a MAC address is [...], it is a UAA.	0
10	If the U/L bit of a MAC address is 0, it is a [...]AA.	U
11	If the U/L bit of a MAC address is [...], it is an LAA.	1
12	If the U/L bit of a MAC address is 1, it is a [...]AA.	L
13	IPv6 [...] addresses are like public IPv4 addresses.	global unicast
14	The IPv6 global unicast range was originally defined as [...]	2000::/3(now it includes all addresses which aren't reserved for other purposes)
15	What are the three sections of an IPv6 global unicast address?[...] Subnet identifier Interface identifier	Global routing prefix
16	What are the three sections of an IPv6 global unicast address?Global routing prefix [...] Interface identifier	Subnet identifier
17	What are the three sections of an IPv6 global unicast address?Global routing prefix Subnet identifier [...]	Interface identifier
18	What is the typical size of the 'global routing prefix' of an IPv6 global unicast address?	48 bits
19	What is the typical size of the 'subnet identifier' of an IPv6 global unicast address?	16 bits
20	What is the typical size of the 'interface identifier' of an IPv6 global unicast address?	64 bits
21	IPv6 [...] addresses are like private IPv4 addresses.	unique local
22	What IPv6 address block is reserved for unique local addresses?	FC00::/7
23	IPv6 unique local addresses begin with [2 hexadecimal digits].	FD
24	Four sections of an IPv6 'unique local' address:[...] Global ID Subnet identifier Interface identifier	Hexadecimal FD (indicates a unique local address)
25	Four sections of an IPv6 'unique local' address:Hexadecimal FD (indicates a unique local address) [...] Subnet identifier Interface identifier	Global ID
26	Four sections of an IPv6 'unique local' address:Hexadecimal FD (indicates a unique local address) Global ID [...] Interface identifier	Subnet identifier

#	Question / Front	Answer / Back
27	Four sections of an IPv6 'unique local' address:Hexadecimal FD (indicates a unique local address) Global ID Subnet identifier [...]	Interface identifier
28	What is the size of the 'global ID' of an IPv6 unique local address?	40 bits
29	Enable IPv6 on an interface without manually configuring an IPv6 address:R1(config-if)# [...]	ipv6 enable
30	What IPv6 address block is reserved for link-local addresses?	FE80::/10
31	How is the interface ID of an IPv6 link-local address generated?	EUI-64
32	IPv6 link-local addresses begin with [four hexadecimal characters]	FE80
33	[...]cast addresses provide one-to-one communication.	Uni
34	[...]cast addresses provide one-to-all communication.	Broad
35	[...]cast addresses provide one-to-many communication.	Multi
36	[...]cast addresses provide one-to-one-of-many communication.	Any
37	What IPv6 address range is reserved for multicast addresses?	FF00::/8
38	Does IPv6 use broadcast messages?	No
39	IPv6 multicast address:[...] = All nodes	FF02::1
40	IPv6 multicast address:FF02::1 = [...]	All nodes
41	IPv6 multicast address:[...] = All routers	FF02::2
42	IPv6 multicast address:FF02::2 = [...]	All routers
43	IPv6 multicast address:[...] = All OSPF routers	FF02::5
44	IPv6 multicast address:FF02::5 = [...]	All OSPF routers
45	IPv6 multicast address:[...] = All OSPF DRs/BDRs	FF02::6
46	IPv6 multicast address:FF02::6 = [...]	All OSPF DRs/BDRs
47	IPv6 multicast address:[...] = All RIP routers	FF02::9
48	IPv6 multicast address:FF02::9 = [...]	All RIP routers
49	IPv6 multicast address:[...] = All EIGRP routers	FF02::A
50	IPv6 multicast address:FF02::A = [...]	All EIGRP routers
51	IPv4 multicast address:[...] = All EIGRP routers	224.0.0.10
52	IPv4 multicast address:224.0.0.10 = [...]	All EIGRP routers
53	IPv4 multicast address:[...] = All RIP routers	224.0.0.9
54	IPv4 multicast address:224.0.0.9 = [...]	All RIP routers
55	IPv4 multicast address:[...] = All OSPF DRs/BDRs	224.0.0.6
56	IPv4 multicast address:224.0.0.6 = [...]	All OSPF DRs/BDRs
57	IPv4 multicast address:[...] = All OSPF routers	224.0.0.5
58	IPv4 multicast address:224.0.0.5 = [...]	All OSPF routers
59	IPv4 multicast address:[...] = All routers	224.0.0.2
60	IPv4 multicast address:224.0.0.2 = [...]	All routers
61	IPv4 multicast address:[...] = All nodes	224.0.0.1
62	IPv4 multicast address:224.0.0.1 = [...]	All nodes
63	IPv6 multicast scope:[...] = Interface-local	FF01
64	IPv6 multicast scope:FF01 = [...]	Interface-local
65	IPv6 multicast scope:[...] = Link-local	FF02
66	IPv6 multicast scope:FF02 = [...]	Link-local
67	IPv6 multicast scope:[...] = Site-local	FF05
68	IPv6 multicast scope:FF05 = [...]	Site-local
69	IPv6 multicast scope:[...] = Organization-local	FF08
70	IPv6 multicast scope:FF08 = [...]	Organization-local
71	IPv6 multicast scope:[...] = Global	FF0E
72	IPv6 multicast scope:FF0E = [...]	Global
73	Configure an IPv6 anycast address:R1(config-if)# [...]	ipv6 address address/prefix-length anycast
74	What is the IPv6 unspecified address?	::
75	What is the IPv6 loopback address?	::1

Day 33 Flashcards - IPv6 (Part 3)

45 flashcards in this section

#	Question / Front	Answer / Back
1	Which IPv6 header field indicates the version of IP?	Version
2	Which IPv6 header field is used for QoS?	Traffic Class
3	Which IPv6 header field is used to identify specific traffic flows?	Flow Label
4	Which IPv6 header field indicates the length of the encapsulated Layer 4 segment?	Payload Length
5	What is the length of the IPv6 header?	40 bytes
6	Which IPv6 header field indicates the encapsulated Layer 4 protocol?	Next Header
7	Which IPv6 header field functions like the IPv4 TTL field?	Hop Limit
8	How is an IPv6 solicited-node multicast address calculated from a unicast address?[...] + Last 6 hex digits of unicast address	ff02::1:ff
9	How is an IPv6 solicited-node multicast address calculated from a unicast address?ff02::1:ff + [...]	Last 6 hex digits of unicast address
10	What does NDP stand for?	Neighbor Discovery Protocol
11	Which protocol replaces ARP in IPv6?	NDP
12	What does the NDP 'NS' message stand for?	Neighbor Solicitation
13	What does the NDP 'NA' message stand for?	Neighbor Advertisement
14	What does the NDP 'RS' message stand for?	Router Solicitation
15	What does the NDP 'RA' message stand for?	Router Advertisement
16	NDP [...] = ICMPv6 Type 135	NS
17	NDP NS = ICMPv6 Type [...]	135
18	NDP [...] = ICMPv6 Type 136	
19	NDP NA = ICMPv6 Type [...]	136
20	NDP [...] = ICMPv6 Type 133	RS
21	NDP RS = ICMPv6 Type [...]	133
22	NDP [...] = ICMPv6 Type 134	RA
23	NDP RA = ICMPv6 Type [...]	134
24	When trying to learn a MAC address, what destination IP address are NDP NS messages sent to?	The neighbor's solicited-node multicast address
25	What command is used to view the IPv6 equivalent of an ARP table?R1# [...]	show ipv6 neighbor
26	NDP RS messages are sent to [destination IP]	FF02::2
27	NDP RA messages are sent to [destination IP]	FF02::1
28	What does SLAAC stand for?	Stateless Address Auto-configuration
29	Which NDP messages allow a device to learn the network prefix for SLAAC?	RS/RA
30	When using SLAAC, how is the interface ID generated?	EUI-64, or randomly
31	What does DAD stand for?	Duplicate Address Detection
32	[...] allows hosts to check if other devices on the local link are using the same IPv6 address.	Duplicate Address Detection (DAD)
33	Which NDP messages does DAD use?	NS/NA
34	To perform DAD, which address does a device send an NS to?	Its own solicited-node multicast address
35	Are routes for IPv6 link-local addresses added to the routing table?	No
36	In a directly attached static route, the [...] is specified.	exit interface (only)
37	In a [...] static route, the exit interface (only) is specified.	directly attached
38	In a [...] static route, the next hop (only) is specified.	recursive
39	In a recursive static route, the [...] is specified.	next hop (only)
40	In a [...] static route, the exit interface and next hop are both specified.	fully specified
41	Configure an IPv6 directly attached static route:R1(config)# [...]	ipv6 route destination/prefix-length exit-interface
42	Configure an IPv6 recursive static route:R1(config)# [...]	ipv6 route destination/prefix-length next-hop
43	Configure an IPv6 fully specified static route:R1(config)# [...]	ipv6 route destination/prefix-length exit-interface next-hop
44	In IPv6, you can't use [...] static routes if the interface is an Ethernet interface.	directly attached(the command will work, but the route will not)
45	To use an IPv6 link-local address as a next hop, you must configure a [...] static route.	fully specified

Day 34 Flashcards - Standard ACLs

22 flashcards in this section

#	Question / Front	Answer / Back
1	What does ACL stand for?	Access Control List

#	Question / Front	Answer / Back
2	ACLs are an ordered sequence of [...]	ACEs (Access Control Entries)
3	ACLs must be [...] to take effect.	applied to an interface
4	ACLs can be applied to an interface [...] or [...]	inbound / outbound
5	In what order do routers check the ACEs of an ACL?	Top to bottom
6	When a packet matches an ACE in an ACL, what happens to the remaining entries below the matching entry?	They are ignored (not checked/processed)
7	How many ACLs can be applied to a single interface?	Two: one inbound one outbound
8	What will happen if a packet doesn't match any of the entries in an ACL?	It will be dropped('implicit deny')
9	The [...] tells the router to deny all traffic that doesn't match any of the configured entries in the ACL.	implicit deny
10	[...] IP ACLs can match based on source IP address only.	Standard
11	[...] IP ACLs can match based on source/destination IP address and source/destination Layer 4 port.	Extended
12	What ranges of numbers can be used to identify standard numbered ACLs?	1-99, 1300-1999
13	Configure a standard numbered ACL entry, specifying the IP/mask.R1(config)# [...]	access-list number {deny permit} ip wildcard-mask
14	Configure a standard numbered ACL entry, permitting or denying all source IPs.R1(config)# [...]	access-list number {deny permit} any
15	Configure a remark for standard numbered ACL:R1(config)# [...]	access-list number remark remark
16	View all ACLs on the router:R1# [...]	show access-lists
17	View all IP ACLs on the router:R1# [...]	show ip access-lists
18	Standard ACLs should be applied as close to the [...] as possible.	destination
19	[...] ACLs should be applied as close to the destination as possible.	Standard
20	Apply an ACL to an interface:R1(config-if)# [...]	ip access-group acl {in out}
21	Enter standard named ACL config mode:R1(config)# [...]	ip access-list standard acl-name
22	Configure a permit or deny entry for a standard named ACL:R1(config-std-nacl)# [...]	[entry-number] {deny permit} ip wildcard-mask

Day 35 Flashcards - Extended ACLs

26 flashcards in this section

#	Question / Front	Answer / Back
1	Delete an ACL entry by specifying the sequence number:R1(config-std-nacl)# [...]	no sequence-number
2	You [can/can't] delete individual ACL entries in global config mode.	can't
3	You [can/can't] delete individual ACL entries in named ACL config mode.	can
4	Resequence an ACL:R1(config)# [...]	ip access-list resequence acl-id starting-seq-num increment
5	Extended numbered ACL ranges:	100-199, 2000-2699
6	Configure an extended ACL entry, specifying protocol, source IP, and destination IP:R1(config)#	access-list number {permit deny} protocol src-ip dest-ip (use either host before each IP address for /32, or specify a wildcard mask)
7	Configure an extended ACL entry, specifying protocol, source IP, and destination IP:R1(config-ext-nacl)#	{permit deny} protocol src-ip dest-ip (use either host before each IP address for /32, or specify a wildcard mask)
8	Enter extended named ACL config mode:R1(config)# [...]	ip access-list extended {name number}
9	[...] = IP protocol number 1	ICMP
10	ICMP = IP protocol number [...]	1
11	[...] = IP protocol number 6	TCP
12	TCP = IP protocol number [...]	6
13	[...] = IP protocol number 17	UDP
14	UDP = IP protocol number [...]	17
15	[...] = IP protocol number 88	EIGRP
16	EIGRP = IP protocol number [...]	88
17	[...] = IP protocol number 89	OSPF
18	OSPF = IP protocol number [...]	89
19	Which extended ACL entry command option?[...] matches a single port.	eq port-num

#	Question / Front	Answer / Back
20	Which extended ACL entry command option?[...] matches all ports greater than the specified number.	gt port-num
21	Which extended ACL entry command option?[...] matches all ports less than the specified number.	lt port-num
22	Which extended ACL entry command option?[...] matches all ports except the specified number.	neq port-num
23	Which extended ACL entry command option?[...] matches the specified range of ports.	range lowest-number highest-number
24	Which command can be used to view which ACLs are applied to an interface?R1# [...]	show ip interface interface-id
25	What is the command to create an extended ACL entry, specifying the protocol, source/destination IP, and source/destination port?R1(config-ext-nacl)# [...]	{permit deny} protocol src-ip src-port dst-ip dst-port *src-ip and dst-ip need the host keyword before or a wildcard mask after *src-port and dst-port need a keyword such as eq before
26	Create an extended ACL entry to permit or deny all traffic:R1(config-ext-nacl)# [...]	{permit deny} ip any any

Day 36 Flashcards - CDP & LLDP

40 flashcards in this section

#	Question / Front	Answer / Back
1	Which Layer 2 discovery protocol is Cisco proprietary?	CDP
2	Which Layer 2 discovery protocol is an industry standard?	LLDP
3	What does CDP stand for?	Cisco Discovery Protocol
4	What does LLDP stand for?	Link Layer Discovery Protocol
5	CDP messages are sent to MAC address [...]	0100.0000.0000
6	Default CDP message timer:	60 seconds
7	Default CDP holdtime:	180 seconds
8	Default CDP version:	Version 2
9	Show global CDP information:R1# [...]	show cdp
10	Show the number of CDP packets sent/received:R1# [...]	show cdp traffic
11	Show the interfaces CDP is enabled on:R1# [...]	show cdp interface
12	Show the CDP neighbor table:R1# [...]	show cdp neighbors
13	Show detailed information for all CDP neighbors:R1# [...]	show cdp neighbors detail
14	CDP [can/can't] share VTP information.	can
15	LLDP [can/can't] share VTP information.	can't
16	Show detailed information for a specific CDP neighbor:R1# [...]	show cdp entry host-name
17	Enable CDP globally:R1(config)# [...]	cdp run
18	Enable CDP on an interface:R1(config-if)# [...]	cdp enable
19	Configure the CDP message timer:R1(config)# [...]	cdp timer seconds
20	Configure the CDP hold time:R1(config)# [...]	cdp holdtime seconds
21	Enable CDPv2: R1(config)# [...]	cdp advertise-v2
22	CDP is [...] on Cisco devices by default.	enabled
23	LLDP is usually [...] on Cisco devices by default.	disabled
24	A device [can/can't] run CDP and LLDP at the same time.	can
25	LLDP messages are sent to MAC address [...]	0180.C200.000E
26	Default LLDP message timer:	30 seconds
27	Default LLDP holdtime:	120 seconds
28	Default LLDP reinitialization timer:	2 seconds
29	Enable LLDP globally:R1(config)# [...]	lldp run
30	Enable LLDP on an interface (tx):R1(config-if)# [...]	lldp transmit
31	Enable LLDP on an interface (rx):R1(config-if)# [...]	lldp receive
32	Configure the LLDP message timer:R1(config)# [...]	lldp timer seconds
33	Configure the LLDP hold time:R1(config)# [...]	lldp holdtime seconds
34	Configure the LLDP reinitialization timer:R1(config)# [...]	lldp reinit seconds
35	Show global LLDP information:R1# [...]	show lldp
36	Show information about LLDP frames sent/received:R1# [...]	show lldp traffic
37	Show LLDP-enabled interfaces:R1# [...]	show lldp interface

#	Question / Front	Answer / Back
38	Show the LLDP neighbor table:R1# [...]	show lldp neighbors
39	Show detailed information about all LLDP neighbors:R1# [...]	show lldp neighbors detail
40	Show detailed information about a single LLDP neighbor:R1# [...]	show lldp entry host-name

Day 37 Flashcards - NTP

39 flashcards in this section

#	Question / Front	Answer / Back
1	Display the software clock:R1# [...]	show clock
2	Display the software clock, including the time source:R1# [...]	show clock detail
3	Display the hardware clock:R1# [...]	show calendar
4	What is the meaning of an asterisk (*) next to the time in the output of show clock?	The time is not authoritative
5	What is the default time zone on Cisco devices?	UTC (Coordinated Universal Time)
6	What does NTP stand for?	Network Time Protocol
7	What is the default time source for the software clock of a Cisco device?	hardware calendar
8	Displays the logs of the device:R1# [...]	show logging
9	Configure the software clock:R1# [...]	clock set hh:mm:ss day month year(day and month can be reversed in order)
10	Configure the hardware clock:R1# [...]	calendar set hh:mm:ss day month year(day and month can be reversed in order)
11	Make the software clock update to match the hardware clock's time:R1# [...]	clock read-calendar
12	Make the hardware clock update to match the software clock's time:R1# [...]	clock update-calendar
13	Configure the timezone:R1(config)# [...]	clock timezone name hours-offset [minutes-offset]
14	Configure recurring daylight saving time:R1(config)# [...]	clock summer-time name recurring start end [offset](start and end require the week (first, last, or number), weekday, month, and time)
15	A device [can/can't] be an NTP server and client at the same time.	can
16	The distance of an NTP server from the original reference clock is called [...].	stratum
17	[...] = UDP port 123	NTP
18	NTP = [...] port 123	UDP
19	NTP = UDP port [...]	123
20	Reference clocks are NTP stratum [...]	0
21	NTP servers directly connected to reference clocks are stratum [...]	1
22	NTP servers which get their time directly from reference clocks are also called [...].	primary servers
23	NTP servers which get their time from other NTP servers are called [...].	secondary servers
24	Anything above NTP stratum [...] is considered unreliable and the device will not sync its time.	15
25	When an NTP server peers with another server at the same stratum level, this is called [...] mode.	symmetric active
26	Configure the NTP server to sync to:R1(config)# [...]	ntp server ip-address [prefer]
27	Show a list of configured NTP servers:R1# [...]	show ntp associations
28	Configure the router to update the hardware clock using NTP:R1(config)# [...]	ntp update-calendar
29	NTP uses only the [...] time zone.	UTC
30	Configure the source interface of NTP messages:R1(config)# [...]	ntp source interface
31	Configure the device to be an NTP server:R1(config)# [...]	ntp master [stratum]
32	The default stratum of the ntp master command is [...].	8
33	Configure NTP symmetric active mode:R1(config)# [...]	ntp peer ip-address
34	Enable NTP authentication:R1(config)# [...]	ntp authenticate
35	Create an NTP authentication key:R1(config)# [...]	ntp authentication-key key-number md5 key
36	Specify an NTP trusted key:R1(config)# [...]	ntp trusted-key key-number
37	Specify the NTP authentication key to use with the NTP server:R1(config)# [...]	ntp server ip-address key key-number
38	Display the status of NTP on the device:R1# [...]	show ntp status
39	NTP servers with a [lower/higher] stratum are preferred.	lower

Day 38 Flashcards - DNS

20 flashcards in this section

#	Question / Front	Answer / Back
1	What does DNS stand for?	Domain Name System
2	Hosts can automatically learn the address of the DNS server via [protocol].	DHCP
3	(Windows) Show basic IP configuration information.> [...]	ipconfig
4	(Windows) Show detailed IP configuration information.> [...]	ipconfig /all
5	(Windows) Show the DNS cache.> [...]	ipconfig /displaydns
6	(Windows) Clear the DNS cache.> [...]	ipconfig /flushdns
7	(Windows) Ping the specified address, specifying the number of pings to send.> [...]	ping ip-address -n number
8	Configure the router to act as a DNS server.R1(config)# [...]	ip dns server
9	Configure an entry in the router's host table.R1(config)# [...]	ip host hostname ip-address
10	Configure the DNS server the router will send DNS queries to.R1(config)# [...]	ip name-server ip-address
11	Enable the router to perform DNS queries. (enable by default) R1(config)# [...]	ip domain lookup(old version: ip domain-lookup)
12	Configure the domain of the router.R1(config)# [...]	ip domain name domain-name(old version: ip domain-name domain-name)
13	Display the router's host table:R1# [...]	show hosts
14	DNS '['...']' records map names to IPv4 addresses.	A
15	DNS 'A' records map names to [...] addresses.	IPv4
16	DNS '['...']' records map names to IPv6 addresses.	AAAA
17	DNS 'AAAA' records map names to [...] addresses.	IPv6
18	Standard DNS queries/responses typically use [TCP/UDP].	UDP
19	[...] is used for DNS messages greater than 512 bytes.	TCP
20	TCP is used for DNS messages greater than [size].	512 bytes

Day 39 Flashcards - DHCP

26 flashcards in this section

#	Question / Front	Answer / Back
1	What does DHCP stand for?	Dynamic Host Configuration Protocol
2	Windows command prompt: Release the PC's DHCP-learned IP address> [...]	ipconfig /release
3	Windows command prompt: Get a new IP address via DHCP> [...]	ipconfig /renew
4	Using the command ipconfig /release in the Windows command prompt will send a DHCP [...] message.	Release
5	The DHCP Release message is [broadcast/unicast].	unicast
6	What four DHCP messages are exchanged for a DHCP client to get an IP address?	DiscoverOffer Request Ack
7	In the DHCP 'DORA' exchange, which messages are sent from client to server?	Discover, Request
8	In the DHCP 'DORA' exchange, which messages are sent from server to client?	Offer, Ack
9	The DHCP Discover message is [broadcast/unicast].	broadcast
10	The DHCP Offer message is [broadcast/unicast].	either
11	The DHCP Request message is [broadcast/unicast].	Broadcast
12	The DHCP Ack message is [broadcast/unicast].	either
13	Which messages in the DHCP 'DORA' exchange are always broadcast?	Discover, Request
14	Which messages in the DHCP 'DORA' exchange can be broadcast or unicast?	Offer, Ack
15	The DHCP Release message is sent from [client/server] to [client/server].	client to server
16	A DHCP relay agent will forward DHCP messages to the DHCP server using [unicast/broadcast].	unicast
17	Configure a range of addresses to be excluded from the DHCP pool.R1(config)# [...]	ip dhcp excluded-address low-address high-address
18	Create a DHCP pool.R1(config)# [...]	ip dhcp pool pool-name
19	Configure the range of addresses for the DHCP pool.R1(dhcp-config)# [...]	network ip-address {/prefix-length subnet-mask}
20	Configure a DNS server for the DHCP pool.R1(dhcp-config)# [...]	dns-server ip-address
21	Configure the domain name for the DHCP pool.R1(dhcp-config)# [...]	domain-name domain-name
22	Configure the default gateway for the DHCP pool.R1(dhcp-config)# [...]	default-router ip-address
23	Configure the lease time for the DHCP pool.R1(dhcp-config)# [...]	lease {days hours minutes infinite}
24	Display the DHCP clients that have received an IP address from this router.R1# [...]	show ip dhcp binding
25	Configure the router as a DHCP relay agent.R1(config-if)# [...]	ip helper-address ip-address

#	Question / Front	Answer / Back
26	Configure the router as a DHCP client on this interface.R1(config-if)# [...]	ip address dhcp

Day 40 Flashcards - SNMP

33 flashcards in this section

#	Question / Front	Answer / Back
1	What does SNMP stand for?	Simple Network Management Protocol
2	What are the two types of devices in SNMP?	Managed DevicesNetwork Management Station (NMS)
3	What does NMS stand for?	Network Management Station
4	The SNMP Manager is software on the [NMS/managed devices].	NMS
5	The SNMP Agent is software on the [NMS/managed devices].	managed devices
6	The [...] is the structure that contains the variables that are managed by SNMP.	Management Information Base (MIB)
7	What does MIB stand for?	Management Information Base
8	What does OID stand for?	Object ID
9	Each variable in the SNMP MIB is identified with an [...].	OID
10	Which version of SNMP is most secure?	SNMPv3
11	What is the most commonly used version of SNMPv2?	SNMPv2c
12	Which SNMP message(s) is/are in the 'read' message class?	GetGetNext GetBulk
13	Which SNMP message(s) is/are in the 'write' message class?	Set
14	Which SNMP message(s) is/are in the 'notification' message class?	TrapInform
15	Which SNMP message(s) is/are in the 'response' message class?	Response
16	Which SNMP message is a request sent from the manager to the agent to retrieve the value of a variable (OID), or multiple variables?	Get
17	Which SNMP message is a request sent from the manager to the agent to discover the available variables in the MIB?	GetNext(also GetBulk, which is a more efficient version)
18	Which SNMP message is a more efficient version of the GetNext message?	GetBulk
19	Which SNMP message is a request sent from the manager to the agent to change the value of one or more variables?	Set
20	Which SNMP message is an unacknowledged notification sent from the agent to the manager?	Trap
21	Which SNMP message is an acknowledged notification sent from the agent to the manager?	Inform
22	SNMP [...] use UDP port 161.	Agents
23	SNMP Agents use [...] port 161.	UDP
24	SNMP Agents use UDP port [...].	161
25	SNMP [...] use UDP port 162.	Managers
26	SNMP Managers use [...] port 162.	UDP
27	SNMP Managers use UDP port [...].	162
28	SNMP: Configure contact information for the device:R1(config)# [...]	snmp-server contact contact-info
29	SNMP: Configure location information for the device:R1(config)# [...]	snmp-server location location
30	SNMP: Configure a read-only community string:R1(config)# [...]	snmp-server community password ro
31	SNMP: Configure a read-write community string:R1(config)# [...]	snmp-server community password rw
32	SNMP: Configure the NMS address, specifying version 2c and a community string.R1(config)# [...]	snmp-server host ip-address version 2c community-string
33	SNMP: Configure the Trap types to send to the NMS:R1(config)# [...]	snmp-server enable traps trap-types

Day 41 Flashcards - Syslog

40 flashcards in this section

#	Question / Front	Answer / Back
1	Syslog message format:[...]:timestamp: %facility-severity-mnemonic:description	seq
2	Syslog message format:seq:[...]: %facility-severity-mnemonic:description	timestamp
3	Syslog message format:seq:timestamp: %[...]severity-mnemonic:description	facility
4	Syslog message format:seq:timestamp: %facility-[...]mnemonic:description	severity
5	Syslog message format:seq:timestamp: %facility-severity-[...]:description	mnemonic
6	Syslog message format:seq:timestamp: %facility-severity-mnemonic:[...]	description
7	Which Syslog message field?-A number that indicates the order of messages.	seq
8	Which Syslog message field?-Indicates the time the message was generated.	time stamp

#	Question / Front	Answer / Back
9	Which Syslog message field?-Indicates which process on the device generated the message.	facility
10	Which Syslog message field?-Indicates how serious the logged event is.	severity
11	Which Syslog message field?-A short code for the message, indicating what happened.	mnemonic
12	Which Syslog message field?-Detailed information about the event being reported.	description
13	Syslog severity level:[...] = Emergency	0
14	Syslog severity level:0 = [...]	Emergency
15	Syslog severity level:[...] = Alert	1
16	Syslog severity level:1 = [...]	Alert
17	Syslog severity level:[...] = Critical	2
18	Syslog severity level:2 = [...]	Critical
19	Syslog severity level:[...] = Error	3
20	Syslog severity level:3 = [...]	Error
21	Syslog severity level:[...] = Warning	4
22	Syslog severity level:4 = [...]	Warning
23	Syslog severity level:[...] = Notice/Notification	5
24	Syslog severity level:5 = [...]	Notice/Notification
25	Syslog severity level:[...] = Informational	6
26	Syslog severity level:6 = [...]	Informational
27	Syslog severity level:[...] = Debugging	7
28	Syslog severity level:7 = [...]	Debugging
29	Syslog servers listen for messages on [...] port 514.	UDP
30	Syslog servers listen for messages on UDP port [...].	514
31	Which locations are Syslog messages sent to by default?	Console lineBuffer
32	Configure Syslog monitoring to the console line, specifying the level.R1(config)# [...]	logging console level
33	Configure Syslog monitoring to the VTY lines, specifying the level.R1(config)# [...]	logging monitor level
34	Configure Syslog monitoring to the buffer, specifying the size/level.R1(config)# [...]	logging buffered [size] level
35	Configure Syslog monitoring to an external serverR1(config)# [...]	logging ip-addressOR logging host ip-address
36	Configure the level of Syslog messages sent to an external server:R1(config)# [...]	logging trap level
37	When connected via Telnet/SSH, display Syslog messages for the current session:R1# [...]	terminal monitor
38	Configure synchronous logging on the device:R1(config-line)# [...]	logging synchronous
39	Enable timestamps for Syslog log messages, displaying the date and time.R1(config)# [...]	service timestamps log datetime
40	Enable sequence numbers for Syslog messages:R1(config)# [...]	service sequence-numbers

Day 42 Flashcards - SSH

28 flashcards in this section

#	Question / Front	Answer / Back
1	Enter console line configuration mode:R1(config)# [...]	line console 0
2	Configure a line password:R1(config-line)# [...]	password password
3	Configure the router to authenticate using the configured line password:R1(config-line)# [...]	login
4	Configure local line authentication:R1(config-line)# [...]	login local
5	By default, is a password required to access the CLI via the console port?	No
6	Configure the switch's default gateway:SW1(config)# [...]	ip default-gateway ip-address
7	What does Telnet stand for?	Teletype Network
8	Does Telnet encrypt data?	No
9	Telnet servers listen for traffic on [...] port 23.	TCP
10	Telnet servers listen for traffic on TCP port [...].	23
11	Access all VTY lines:R1(config)# [...]	line vty 0 15
12	Configure the exec timeout:R1(config-line)# [...]	exec-timeout minutes seconds
13	Configure the protocols that can be used to connect via the VTY lines:R1(config-line)# [...]	transport input [protocols all none]
14	Apply an ACL for connections to the VTY lines:R1(config-line)# [...]	access-class acl in
15	Generate an RSA key pair:R1(config)# [...]	crypto key generate rsa
16	A device that supports both SSHv1 and SSHv2 is said to run version [...]	1.99
17	SSH servers listen for traffic on [...] port 22.	TCP
18	SSH servers listen for traffic on TCP port [...].	22

#	Question / Front	Answer / Back
19	Does SSH encrypt traffic?	Yes
20	Show information about the IOS version of the device:R1# [...]	show version
21	Show information about the current SSH configuration:R1# [...]	show ip ssh
22	[...] IOS images support SSH.	K9
23	[...] IOS images do not support cryptographic features like SSH.	NPE (No Payload Encryption)
24	What does FQDN stand for?	Fully Qualified Domain Name
25	What makes up the FQDN of a device?	hostname.domainname
26	When crypto key generate rsa is used, how are the keys named?	The FQDN is used as the name
27	Limit SSH connections to SSHv2 only:R1(config)# [...]	ip ssh version 2
28	What does SSH stand for?	Secure Shell

Day 43 Flashcards - FTP&TFTP;

36 flashcards in this section

#	Question / Front	Answer / Back
1	What does FTP stand for?	File Transfer Protocol
2	What does TFTP stand for?	Trivial File Transfer Protocol
3	Does TFTP use authentication?	No (no username/PW)
4	Does TFTP use encryption?	No
5	TFTP servers listen on UDP port [...]	69
6	TFTP uses '['...' communication, in which the client and server alternately send messages.	lock-step
7	What are the three phases of a TFTP file transfer?	ConnectionData Transfer Connection Termination
8	What does the TFTP 'TID' stand for?	Transfer ID
9	The random ports used in TFTP are also called:	TIDs (Transfer IDs)
10	Which protocol is more lightweight and simple, FTP or TFTP?	TFTP
11	Which protocol provides more functionality, FTP or TFTP?	FTP
12	FTP [...] connections use TCP port 21.	control
13	FTP control connections use [...] port 21.	TCP
14	FTP control connections use TCP port [...].	21
15	FTP [...] connections use TCP port 20.	data
16	FTP data connections use [...] port 20.	TCP
17	FTP data connections use TCP port [...].	20
18	TFTP servers listen on [...] port 69.	UDP
19	In FTP [...] mode, the server initiates the data connection.	active
20	In FTP active mode, the [...] initiates the data connection.	server
21	In FTP [...] mode, the client initiates the data connection.	passive
22	In FTP passive mode, the [...] initiates the data connection.	client
23	The [...] file system type is used for storage devices such as flash memory.	disk
24	The [...] file system type represents logical systems used for internal functions.	opaque
25	The [...] file system type is used for the device's internal NVRAM.	nvrn
26	The [...] file system type represents external file systems, for example FTP/TFTP servers.	network
27	Display the contents of flash memory:R1# [...]	show flash
28	Display the file systems of the device:R1# [...]	show file systems
29	Specify the OS file the device should use when next time it boots:R1(config)# [...]	boot system filepath
30	Copy a file from a TFTP server to the router's flash:R1# [...]	copy tftp: flash:
31	Copy a file from an FTP server to the router's flash:R1# [...]	copy ftp: flash:
32	Configure the FTP username of the device:R1(config)# [...]	ip ftp username username
33	Configure the FTP password of the device:R1(config)# [...]	ip ftp password password
34	Does FTP use authentication?	Yes (username/PW)
35	Does FTP use encryption?	No(FTPS does)
36	What two kinds of connection are used in FTP?	ControlData

Day 44 Flashcards - NAT (Part 1)

19 flashcards in this section

#	Question / Front	Answer / Back
1	Private IPv4 addresses are defined in RFC [...].	1918
2	What is the class A range of private IPv4 addresses?	10.0.0.0/8
3	What is the class B range of private IPv4 addresses?	172.16.0.0/12
4	What is the class C range of private IPv4 addresses?	192.168.0.0/16
5	Can private IPv4 addresses be used over the Internet?	No
6	What does NAT stand for?	Network Address Translation
7	[...] NAT involves statically configuring one-to-one IP address mappings.	Static
8	Static Source NAT = [...] address is mapped to [...] address.	Inside Local / Inside Global
9	Cisco NAT terminology:[...] = The IP address of the inside host from the perspective of the inside network.	Inside Local
10	Cisco NAT terminology:[...] = The IP address of the inside host from the perspective of the outside network.	Inside Global
11	Cisco NAT terminology:[...] = The IP address of the outside host from the perspective of the inside network.	Outside Local
12	Cisco NAT terminology:[...] = The IP address of the outside host from the perspective of the outside network.	Outside Global
13	Configure the NAT inside interface:R1(config-if)# [...]	ip nat inside
14	Configure the NAT outside interface:R1(config-if)# [...]	ip nat outside
15	Configure a static source NAT mapping:R1(config)# [...]	ip nat inside source static inside-local-ip inside-global-ip
16	When only source NAT is used, in the output of show ip nat translations you can expect the [...] and [...] addresses to be the same.	outside local / outside global
17	Show the NAT translation table:R1# [...]	show ip nat translations
18	Show the NAT statistics:R1# [...]	show ip nat statistics
19	Clear all dynamic NAT translation entries:R1# [...]	clear ip nat translation *

Day 45 Flashcards - NAT (Part 2)

14 flashcards in this section

#	Question / Front	Answer / Back
1	Configure a pool for dynamic NAT:R1(config)# [...]	ip nat pool pool-name start-ip end-ip [prefix-length prefix-length netmask subnet-mask]
2	Configure dynamic NAT:R1(config)# [...]	ip nat inside source list access-list pool pool-name
3	Configure dynamic PAT using an address pool:R1(config)# [...]	ip nat inside source list access-list pool pool-name overload
4	Configure dynamic PAT using the router's interface IP:R1(config)# [...]	ip nat inside source list access-list interface interface overload
5	In dynamic NAT/PAT, what is used to identify traffic that should be translated?	an ACL
6	In dynamic NAT, what is used to identify the range of public IP address that can be used for translations?	A NAT pool
7	What does PAT stand for?	Port Address Translation
8	What is another name for PAT?	NAT Overload
9	Which form of NAT is most useful for preserving public IP addresses?	PAT / NAT Overload
10	What does a router do if a packet requires NAT but no inside global addresses are available?	It drops the packet
11	When an ACL is used to identify traffic in dynamic NAT/PAT, what happens to traffic permitted by the ACL?	It is translated by the router
12	When an ACL is used to identify traffic in dynamic NAT/PAT, what happens to traffic denied by the ACL?	It is not translated by the router (but the traffic is not dropped!)
13	In which form of NAT does the router automatically create one-to-one IP address mappings?	Dynamic NAT
14	In which form of NAT are multiple inside local addresses translated to a single inside global address at the same time?	PAT / NAT Overload

Day 46 Flashcards - QoS (Part 1)

41 flashcards in this section

#	Question / Front	Answer / Back
1	What does PSTN stand for?	Public Switched Telephone Network
2	What does POTS stand for?	Plain Old Telephone Service
3	What does VoIP stand for?	Voice over IP

#	Question / Front	Answer / Back
4	When the switchport voice vlan command is applied to an interface, which traffic will be tagged? Traffic from the PC or from the IP phone?	Traffic from the IP phone (traffic from the PC will be untagged as usual for an access port)
5	Configure the voice VLAN:SW1(config-if)# [...]	switchport voice vlan vlan-number
6	What does PoE stand for?	Power over Ethernet
7	PoE: What does PSE stand for?	Power Sourcing Equipment
8	PoE: What does PD stand for?	Powered Device
9	Electronic devices use [AC/DC] power.	DC
10	PoE: [...] can be configured to prevent a PD from taking too much power.	Power policing
11	Power policing: Configure the switch to disable the interface and generate a Syslog message if the PD draws too much power:SW1(config-if)# [...]	power inline police power inline police action errdisable
12	Power policing: Configure the switch to restart the interface and generate a Syslog message if the PD draws too much power:SW1(config-if)# [...]	power inline police action log
13	What does Cisco ILP stand for?	Cisco Inline Power
14	PoE (Type 1) is IEEE [...]	802.3af
15	PoE+ (Type 2) is IEEE [...]	802.3at
16	UPoE (Type 3) is IEEE [...]	802.3bt
17	UPoE+ (Type 4) is IEEE [...]	802.3bt
18	PoE (Type 1) offers [...] watts of power.	15
19	PoE+ (Type 2) offers [...] watts of power.	30
20	UPoE (Type 3) offers [...] watts of power.	60
21	UPoE+ (Type 4) offers [...] watts of power.	100
22	Cisco ILP offers [...] watts of power.	7
23	Cisco ILP uses [...] powered wire pairs.	2
24	PoE (Type 1) uses [...] powered wire pairs.	2
25	PoE+ (Type 2) uses [...] powered wire pairs.	2
26	UPoE (Type 3) uses [...] powered wire pairs.	4
27	UPoE+ (Type 4) uses [...] powered wire pairs.	4
28	QoS: [...] refers to the overall capacity of a link.	Bandwidth
29	QoS: [...] is the amount of time it takes traffic to go from source to destination.	Delay (=one-way delay) (Two-way delay measures both ways)
30	QoS: [...] is the variation in one-way delay between packets sent by the same application.	Jitter
31	QoS: [...] is the percentage of packets that do not reach their destination.	Loss
32	What is considered acceptable one-way delay for interactive audio traffic?	150 ms or less
33	What is considered acceptable jitter for interactive audio traffic?	30 ms or less
34	What is considered acceptable loss for interactive audio traffic?	1% or less
35	QoS: By default, queued messages will be forward in a [...] manner.	FIFO
36	What does FIFO stand for?	First In First Out
37	When packets are dropped due to a full queue, it is called [...].	tail drop
38	Tail drop is harmful because it can lead to [...].	TCP global synchronization
39	QoS: What does RED stand for?	Random Early Detection
40	QoS: What does WRED stand for?	Weighted Random Early Detection
41	QoS: What features drop random packets when traffic queues pass a certain threshold?	RED / WRED

Day 47 Flashcards - QoS (Part 2)

55 flashcards in this section

#	Question / Front	Answer / Back
1	What does NBAR stand for?	Network Based Application Recognition
2	What Layer 2 header field can be used for QoS markings?	PCP / CoS
3	What Layer 3 header field can be used for QoS markings?	DSCP(formerly IPP)
4	What does IPP stand for?	IP Precedence
5	[...] delivery means there is no guarantee that data is delivered or that it meets any QoS standard.	Best effort
6	CoS/PCP [...] = best effort	0
7	CoS/PCP 0 = [...]	best effort
8	CoS/PCP [...] = critical applications	3

#	Question / Front	Answer / Back
9	CoS/PCP 3 = [...]	critical applications
10	CoS/PCP [...] = video traffic	4
11	CoS/PCP 4 = [...]	video traffic
12	CoS/PCP [...] = voice traffic	5
13	CoS/PCP 5 = [...]	voice traffic
14	The IPP precedence field was [...] in length.	3 bits
15	DSCP: What does DF stand for?	default forwarding
16	DSCP: What does EF stand for?	expedited forwarding
17	DSCP: What does AF stand for?	assured forwarding
18	DSCP: What does CS stand for?	class selector
19	[...] = DSCP 0	DF
20	DF = DSCP [...]	0
21	[...] = DSCP 46	EF
22	EF = DSCP [...]	46
23	AF41 = DSCP [...]	34
24	AF42 = DSCP [...]	36
25	AF43 = DSCP [...]	38
26	AF31 = DSCP [...]	26
27	AF32 = DSCP [...]	28
28	AF33 = DSCP [...]	30
29	AF21 = DSCP [...]	18
30	AF22 = DSCP [...]	20
31	AF23 = DSCP [...]	22
32	AF11 = DSCP [...]	10
33	AF12 = DSCP [...]	12
34	AF13 = DSCP [...]	14
35	CS0 = DSCP [...]	0
36	CS1 = DSCP [...]	8
37	CS2 = DSCP [...]	16
38	CS3 = DSCP [...]	24
39	CS4 = DSCP [...]	32
40	CS5 = DSCP [...]	40
41	CS6 = DSCP [...]	48
42	CS7 = DSCP [...]	56
43	RFC 4594 recommends the [...] marking for voice traffic.	EF
44	RFC 4594 recommends the [...] marking for interactive video traffic.	AF4x (AF41, AF42, AF43)
45	RFC 4594 recommends the [...] marking for streaming video traffic.	AF3x (AF31, AF32, AF33)
46	RFC 4594 recommends the [...] marking for high priority data traffic.	AF2x (AF21, AF22, AF23)
47	RFC 4594 recommends the [...] marking for best effort traffic.	DF
48	If an IP phone is connected to a switchport, the QoS trust boundary should be at the [...].	IP phone
49	What does CBWFQ stand for?	Class-Based Weighted Fair Queuing
50	What QoS scheduling algorithm uses a weighted round-robin scheduler while guaranteeing a minimum bandwidth to each queue?	CBWFQ
51	What does LLQ stand for?	Low Latency Queuing
52	LLQ designates one or more queues as [...] queues.	strict priority
53	[...] buffers traffic in a queue if the traffic rate goes over the configured rate.	Shaping
54	[...] drops traffic in a queue if the traffic rate goes over the configured rate.	Policing
55	IP phones mark their call signaling traffic as PCP/CoS [...].	3

Day 48 Flashcards - Security Fundamentals

54 flashcards in this section

#	Question / Front	Answer / Back
1	What is the C of the CIA triad?	Confidentiality
2	What is the I of the CIA triad?	Integrity
3	What is the A of the CIA triad?	Availability

#	Question / Front	Answer / Back
4	CIA Triad: [...] = only authorized users should be able to access data.	Confidentiality
5	CIA Triad: [...] = data should not be tampered with by unauthorized users.	Integrity
6	CIA Triad: [...] = the network/systems should be operational and accessible to authorized users.	Availability
7	A [...] is any potential weakness that can compromise the security of a system.	vulnerability
8	An [...] is something that can potentially be used to take advantage of a vulnerability.	exploit
9	A [...] is the potential of a vulnerability to be exploited.	threat
10	A [...] technique is something that can protect against threats.	mitigation
11	What does DoS attack stand for?	denial-of-service
12	What does DDoS attack stand for?	distributed denial-of-service
13	A TCP [...] attack exploits the TCP three-way handshake.	SYN flood
14	A TCP SYN flood results in a [...] on the target.	denial-of-service
15	To [...] an address is to use a fake source address (IP/MAC)	spoof
16	In a [...] attack, an attacker floods the target with DHCP Discover messages.	DHCP exhaustion
17	In a [...] attack, an attacker causes a reflector to send traffic to the target.	reflection
18	An [...] attack is a more dangerous kind of reflection attack.	amplification
19	In a [...] attack, the attacker intercepts communications between two devices.	man-in-the-middle
20	ARP spoofing is also known as [...].	ARP poisoning
21	In an [...] attack, an attacker sends fraudulent ARP replies.	ARP spoofing
22	[...] attacks are used to gather information about a target.	Reconnaissance
23	[...] refers to a variety of harmful programs that can infect a computer.	Malware
24	[...] are malware that infect other software, called a host program.	Viruses
25	[...] are standalone malware that spread on their own, without user interaction.	Worms
26	[...] are malware that is disguised as legitimate software.	Trojan horses
27	[...] attacks involve psychological manipulation to make the target reveal confidential information or perform some other action.	Social engineering
28	[...] is a kind of social engineering which typically involves fraudulent emails that try to appear legitimate.	Phishing
29	[...] is a more targeted form of phishing, for example aimed at employees of a certain company.	Spear phishing
30	[...] is a kind of phishing targeted at high-profile individuals.	Whaling
31	[...] is phishing performed over the phone.	Vishing
32	[...] is phishing using SMS text messages.	Smishing
33	[...] attacks compromise sites that the target victim frequently visits.	Watering hole
34	[...] attacks involve entering restricted, secured areas by walking in behind an authorized person as they enter.	Tailgating
35	A [...] attack runs through a list of common words or passwords to find the target's password.	dictionary
36	A [...] attack tries every possible combination of letters, numbers, and special characters to guess the target's password.	brute force
37	What does MFA stand for?	Multi-factor authentication
38	What are the three factors of multi-factor authentication?	Something you know Something you have Something you are
39	[...] are a form of authentication often used to prove the identity of a website.	Digital certificates
40	What does AAA stand for?	Authentication, Authorization, and Accounting
41	AAA: [...] is the process of verifying a user's identity.	Authentication
42	AAA: [...] is the process of granting a user the appropriate access and permissions.	Authorization
43	AAA: [...] is the process of recording the user's activities on the system.	Accounting
44	Cisco's AAA server is [...]	ISE
45	What does Cisco ISE stand for?	Identity Services Engine
46	Which AAA protocol is an open standard?	RADIUS
47	Which AAA protocol is Cisco proprietary?	TACACS+
48	RADIUS uses UDP ports [...] and [...].	1812 / 1813
49	RADIUS uses [...] ports 1812 and 1813.	UDP
50	TACACS+ uses TCP port [...]	49
51	TACACS+ uses [...] port 49	TCP
52	[...] programs are designed to make employees aware of potential security threats and risks.	User awareness

#	Question / Front	Answer / Back
53	[...] programs are formal sessions which educate users on corporate security policies and issues.	User training
54	[...] protects equipment and data from potential attackers by only allowing authorized users into protected areas such as network closets.	Physical access control

Day 49 Flashcards - Port Security

34 flashcards in this section

#	Question / Front	Answer / Back
1	A combination of statically-configured and dynamically-learned secure MAC addresses [can/can't] exist on a switchport.	can
2	Enable port security on an interface:SW1(config-if)# [...]	switchport port-security
3	Manually configure a secure MAC address on the interface. SW1(config-if)# [...]	switchport port-security mac-address mac-address
4	Enable sticky secure MAC address learning:SW1(config-if)# [...]	switchport port-security mac-address sticky
5	Configure the port security violation mode:SW1(config-if)# [...]	switchport port-security violation {shutdown restrict protect}
6	Configure the port security aging time:SW1(config-if)# [...]	switchport port-security aging time minutes
7	Configure the port security aging type:SW1(config-if)# [...]	switchport port-security aging type {absolute inactivity}
8	Enable aging of static secure MAC addressesSW1(config-if)# [...]	switchport port-security aging static
9	Enable errdisable recovery for port security violations:SW1(config)# [...]	errdisable recovery cause psecure-violation
10	Configure the errdisable recovery interval:SW1(config)# [...]	errdisable recovery interval seconds
11	Show all secure MAC addresses on the switch:SW1# [...]	show mac address-table secure
12	Show a summary of port security-enabled switchportsSW1# [...]	show port-security
13	Show port security information for an individual switchportSW1# [...]	show port-security interface interface
14	Show a summary of errdisable recovery information:SW1# [...]	show errdisable recovery
15	Port security [can/can't] be enabled on an interface configured as switchport mode access	can
16	Port security [can/can't] be enabled on an interface configured as switchport mode trunk	can
17	Port security [can/can't] be enabled on an interface configured as switchport mode dynamic auto	can't
18	Port security [can/can't] be enabled on an interface configured as switchport mode dynamic desirable	can't
19	When an interface is shutdown by port security, its status in the output of show interfaces status will be [...]	err-disabled
20	Manually re-enable an interface disabled by port security:SW1(config-if)# [...] SW1(config-if)# [...]	shutdownno shutdown
21	The default errdisable recovery timer is [...]	300 seconds (5 minutes)
22	Errdisable recovery is [enabled/disabled] by default for all causes.	disabled
23	What is the default port security violation mode?	Shutdown
24	Port security shutdown violation mode:Syslog/SNMP messages [are/aren't] generated.	are
25	Port security restrict violation mode:Syslog/SNMP messages [are/aren't] generated.	are
26	Port security protect violation mode:Syslog/SNMP messages [are/aren't] generated.	aren't
27	Port security shutdown violation mode:The interface [is/isn't] disabled.	is
28	Port security restrict violation mode:The interface [is/isn't] disabled.	isn't
29	Port security protect violation mode:The interface [is/isn't] disabled.	isn't
30	Port security shutdown violation mode:The violation counter [is/isn't] incremented.	is
31	Port security restrict violation mode:The violation counter [is/isn't] incremented.	is
32	Port security protect violation mode:The violation counter [is/isn't] incremented.	isn't
33	The default port security aging type is [...].	absolute
34	Port security secure static address aging is [enabled/disabled] by default.	disabled

Day 50 Flashcards - DHCP Snooping

21 flashcards in this section

#	Question / Front	Answer / Back
1	Enable DHCP Snooping on the switch:SW1(config)# [...]	ip dhcp snooping
2	Enable DHCP Snooping on a VLAN:SW1(config)# [...]	ip dhcp snooping vlan vlan-number
3	Enable errdisable recovery for DHCP rate limiting violations:SW1(config)# [...]	errdisable recovery cause dhcp-rate-limit

#	Question / Front	Answer / Back
4	Configure the DHCP snooping-enabled switch to not add option 82 to DHCP messages:SW1(config)# [...]	no ip dhcp snooping information option
5	Configure a DHCP snooping trusted port:SW1(config-if)# [...]	ip dhcp snooping trust
6	Limit the rate that DHCP messages are allowed on a DHCP snooping enabled interface:SW1(config-if)# [...]	ip dhcp snooping limit rate packets-per-second
7	Show the DHCP snooping binding table:SW1# [...]	show ip dhcp snooping binding
8	When DHCP snooping is enabled, all interfaces are [trusted/untrusted] by default.	untrusted
9	DHCP Server or DHCP Client message?OFFER	Server
10	DHCP Server or DHCP Client message?ACK	Server
11	DHCP Server or DHCP Client message?NAK	Server
12	DHCP Server or DHCP Client message?DISCOVER	Client
13	DHCP Server or DHCP Client message?REQUEST	Client
14	DHCP Server or DHCP Client message?RELEASE	Client
15	DHCP Server or DHCP Client message?DECLINE	Client
16	DHCP snooping untrusted port:If a DHCP [...] message is received, discard it with no further checks.	Server
17	DHCP snooping untrusted port:If a DHCP [...] message is received, inspect the message to determine if it should be forwarded or discarded.	client
18	DHCP Snooping: When a client successfully leases an IP address from a server, create a new entry in the [...]	DHCP Snooping Binding Table
19	If DHCP snooping rate-limiting is configured and the rate is exceeded, what happens?	The interface is err-disabled
20	DHCP Option [...] is also known as the DHCP Relay Agent Information Option.	82
21	By default, Cisco switches will drop DHCP messages with Option 82 that are received on an [...] port.	untrusted

Day 51 Flashcards - Dynamic ARP Inspection

17 flashcards in this section

#	Question / Front	Answer / Back
1	Enable DAI for a VLAN:SW1(config)# [...]	ip arp inspection vlan vlan
2	Enable err-disable recovery for DAI rate limiting:SW1(config)# [...]	errdisable recovery cause arp-inspection
3	Enable additional DAI validation checks:SW1(config)# [...]	ip arp inspection validate (src-mac dst-mac ip)
4	Configure a DAI trusted interface: SW1(config-if)# [...]	ip arp inspection trust
5	Configure DAI rate limiting: SW1(config-if)# [...]	ip arp inspection limit rate packets [burst interval seconds]
6	Create an ARP ACL: SW1(config)# [...]	arp access-list name
7	Configure an ARP ACL entry mapping an IP address to a MAC address (permit) SW1(config-arp-nacl)# [...]	permit ip host ip-address mac host mac-address
8	Apply an ARP ACL to DAI: SW1(config)# [...]	ip arp inspection filter arp-acl-name vlan vlan
9	Show a summary of DAI configuration and statistics: SW1# [...]	show ip arp inspection
10	Show a summary of DAI interfaces: SW1# [...]	show ip arp inspection interfaces
11	DAI: All ports are [trusted/untrusted] by default.	untrusted
12	DAI checks ARP messages' [...] and [...] fields against the DHCP snooping binding table and ARP ACLs.	sender MAC / sender IP
13	DAI checks ARP messages' sender MAC and sender IP fields against the [...] and [...].	DHCP snooping binding table / ARP ACLs
14	DAI inspects ARP messages received on [...] ports.	untrusted
15	Does DAI inspect messages received on trusted ports?	No
16	DAI rate limiting is enabled on [...] ports by default.	untrusted
17	DAI rate limiting is enabled on untrusted ports with a rate of [...] by default.	15 packets per second

Day 52 Flashcards - LAN Architectures

21 flashcards in this section

#	Question / Front	Answer / Back
1	When several devices all connect to one central device, it is called a [...] topology.	star

#	Question / Front	Answer / Back
2	When each device connects to all other devices, it is called a [...] topology.	full mesh
3	When some devices are connected to each other, but not all, it is called a [...] topology.	partial mesh
4	Which three layers are found in a traditional 3-tier campus LAN?	AccessDistribution Core
5	Which two layers are found in a traditional 2-tier campus LAN?	AccessDistribution
6	The [...] layer of a campus LAN is also known as the aggregation layer.	distribution
7	End hosts connect to the [...] layer of a campus LAN.	access
8	QoS marking is typically done at the [...] layer of a campus LAN.	access
9	PoE is often used at the [...] layer of a campus LAN.	access
10	In a campus LAN, the [...] layer aggregates connections from access layer switches.	distribution
11	In a campus LAN, the [...] layer is typically the border between Layer 2 and Layer 3.	distribution
12	In a large campus LAN, the [...] layer connects distribution layers together.	core
13	Spine-Leaf architecture is efficient in data centers with lots of [...] traffic.	east-west
14	Spine-Leaf Architecture: Every leaf switch is connected to every [...] switch.	spine
15	Spine-Leaf Architecture: Every spine switch is connected to every [...] switch.	leaf
16	Spine-Leaf Architecture: A leaf switch must not connect to a [...] switch.	leaf
17	Spine-Leaf Architecture: A spine switch must not connect to a [...] switch.	spine
18	Spine-Leaf Architecture: End hosts connect to [...] switches.	leaf
19	What does SOHO stand for?	Small Office/Home Office
20	In SOHO networks, all networking functions are typically provided by a multipurpose network device called a [...].	home router / wireless router
21	A 2-tier campus LAN architecture is also known as a [...] architecture.	Collapsed-core

Day 53 Flashcards - WAN Architectures

37 flashcards in this section

#	Question / Front	Answer / Back
1	What leased line standard provides 1.544 Mbps of bandwidth?	T1
2	What leased line standard provides 6.312 Mbps of bandwidth?	T2
3	What leased line standard provides 44.736 Mbps of bandwidth?	T3
4	What leased line standard provides 2.048 Mbps of bandwidth?	E1
5	What leased line standard provides 8.448 Mbps of bandwidth?	E2
6	What leased line standard provides 34.368 Mbps of bandwidth?	E3
7	What does MPLS stand for?	Multi Protocol Label Switching
8	In MPLS, what does CE router stand for?	Customer Edge
9	In MPLS, what does PE router stand for?	Provider Edge
10	In MPLS, what does P router stand for?	Provider (Core)
11	In [...] MPLS VPNs, CE routers and PE routers can form routing protocol peerings.	Layer 3
12	In [...] MPLS VPNs, the entire service provider network is transparent to the customer.	Layer 2
13	In [...] MPLS VPNs, CE routers form routing protocol peerings with each other.	Layer 2
14	What does DSL stand for?	Digital Subscriber Line
15	[...] provides Internet connectivity via phone lines.	DSL
16	[...] provides Internet connectivity via TV lines.	Cable (CATV) Internet
17	What does modem stand for?	modulator-demodulator
18	A DSL [...] is required to convert data into a format suitable to be sent over the phone lines.	modem
19	A cable [...] is required to convert data into a format suitable to be sent over the TV lines.	modem
20	Internet connectivity: 1 connection to 1 ISP = [...]	Single Homed
21	Internet connectivity: 2 connections to 1 ISP = [...]	Dual Homed
22	Internet connectivity: 1 connection to each of 2 ISPs = [...]	Multihomed
23	Internet connectivity: 2 connections to each of 2 ISPs = [...]	Dual Multihomed
24	Site-to-site VPNs typically use [protocol].	IPsec
25	Remote-access VPNs typically use [protocol].	TLS
26	[...] VPNs typically use IPsec.	Site-to-site
27	[...] VPNs typically use TLS.	Remote-access
28	[...] VPNs are used to permanently connect two sites over the Internet.	Site-to-site
29	[...] VPNs are used to provide on-demand access to the company network for end devices.	Remote-access

#	Question / Front	Answer / Back
30	What does GRE stand for?	Generic Routing Encapsulation
31	What does DMVPN stand for?	Dynamic Multipoint VPN
32	[...] is a Cisco-developed solution that allows routers to dynamically create a full mesh of IPsec tunnels without having to manually configure every single tunnel.	DMVPN
33	[...] creates tunnels like IPsec, but does not encrypt the original packet.	GRE
34	Does IPsec support broadcast and multicast traffic?	No
35	Does GRE support broadcast and multicast traffic?	Yes
36	What does TLS stand for?	Transport Layer Security
37	What does SSL stand for?	Secure Sockets Layer

Day 54 (Part 1) Flashcards - Virtualization & Cloud

32 flashcards in this section

#	Question / Front	Answer / Back
1	What does Cisco UCS stand for?	Unified Computing System
2	What does VM stand for?	Virtual Machine
3	A [...] is used to manage and allocate the hardware resources to each VM.	hypervisor
4	Another name for a hypervisor is [...].	VMM (Virtual Machine Monitor)
5	What does VMM stand for?	Virtual Machine Monitor
6	A Type [...] hypervisor runs directly on top of the hardware.	1
7	A Type [...] hypervisor is also called a bare-metal hypervisor.	1
8	A Type [...] hypervisor is also called a native hypervisor.	1
9	A Type [...] hypervisor is also called a hosted hypervisor.	2
10	A Type [...] hypervisor runs as a program on an operating system.	2
11	When using a Type 2 hypervisor, the OS running directly on the hardware is called the [...] OS.	Host
12	When using a Type 2 hypervisor, the OS running in a VM is called a [...] OS.	Guest
13	Five essential characteristics of cloud computing:- [...] - Broad network access - Resource pooling - Rapid elasticity - Measured service	On-demand self-service
14	Five essential characteristics of cloud computing:- On-demand self-service - [...] - Resource pooling - Rapid elasticity - Measured service	Broad network access
15	Five essential characteristics of cloud computing:- On-demand self-service - Broad network access - [...] - Rapid elasticity - Measured service	Resource pooling
16	Five essential characteristics of cloud computing:- On-demand self-service - Broad network access - Resource pooling - [...] - Measured service	Rapid elasticity
17	Five essential characteristics of cloud computing:- On-demand self-service - Broad network access - Resource pooling - Rapid elasticity - [...]	Measured service
18	Three service models of cloud computing (according to NIST):- [...] - Platform as a Service (PaaS) - Infrastructure as a Service (IaaS)	Software as a Service (SaaS)
19	Three service models of cloud computing (according to NIST):- Software as a Service (SaaS) - [...] - Infrastructure as a Service (IaaS)	Platform as a Service (PaaS)
20	Three service models of cloud computing (according to NIST):- Software as a Service (SaaS) - Platform as a Service (PaaS) - [...]	Infrastructure as a Service (IaaS)
21	What does SaaS stand for?	Software as a Service
22	What does PaaS stand for?	Platform as a Service
23	What does IaaS stand for?	Infrastructure as a Service

#	Question / Front	Answer / Back
24	Microsoft Office 365 is an example of [cloud service model].	SaaS
25	AWS Lambda is an example of [cloud service model].	PaaS
26	Google App Engine is an example of [cloud service model].	PaaS
27	Amazon EC2 is an example of [cloud service model].	IaaS
28	Google Compute Engine is an example of [cloud service model].	IaaS
29	Four deployment models of cloud (according to NIST):- [...] - Community cloud - Public cloud - Hybrid cloud	Private cloud
30	Four deployment models of cloud (according to NIST):- Private cloud - [...] - Public cloud - Hybrid cloud	Community cloud
31	Four deployment models of cloud (according to NIST):- Private cloud - Community cloud - [...] - Hybrid cloud	Public cloud
32	Four deployment models of cloud (according to NIST):- Private cloud - Community cloud - Public cloud - [...]	Hybrid cloud

Day 54 (part 2) Flashcards - Containers

13 flashcards in this section

#	Question / Front	Answer / Back
1	Kubernetes is an example of a [software type].	container orchestrator
2	Docker Swarm is an example of a [software type].	container orchestrator
3	Docker Engine is an example of a [software type].	container engine
4	[...] are software packages that contain an app and all dependencies for the app to run.	Containers
5	Containers [do/do not] run an OS in the container.	do not
6	Containers run on top of a [software type].	container engine
7	A container engine is run on a [...].	host OS (ie. Linux)
8	A [...] is a software platform for automating the deployment, management, scaling etc. of containers.	container orchestrator
9	Which boots up faster, a VM or a container?	a container
10	Which takes up more disk space, a VM or a container?	a VM
11	Which uses more CPU/RAM, a VM or a container?	a VM
12	Which is more isolated, a VM or a container?	a VM *because each VM runs its own OS, whereas containers all share the same OS
13	A [VM/container] runs its own OS instance.	VM

Day 54 (Part 3) Flashcards - VRF

9 flashcards in this section

#	Question / Front	Answer / Back
1	What does VRF stand for?	Virtual Routing and Forwarding
2	Only Layer [...] interfaces can be configured in a VRF.	3
3	[...] can be configured to allow traffic to pass between VRFs.	VRF Leaking
4	VRF allows a router to build multiple [...]	routing tables
5	Without VRF, two interfaces on the same router [can/cannot] be in the same subnet.	cannot
6	Create a VRF: R1(config)# [...]	ip vrf vrf-name
7	Assign an interface to a VRF: R1(config-if)# [...]	ip vrf forwarding vrf-name
8	Show the routing table for a VRF: R1# [...]	show ip route vrf vrf-name
9	Ping in a VRF: R1# [...]	ping vrf vrf-name ip-address

Day 55 Flashcards - Wireless Fundamentals

69 flashcards in this section

#	Question / Front	Answer / Back
1	Wireless LANs are defined in IEEE [...].	802.11
2	The [...] tests and certifies equipment for 802.11 standards compliance interoperability with other devices.	Wi-Fi Alliance
3	Wireless networks operate in [...] duplex.	half
4	Wireless networks use [...] to avoid collisions.	CSMA/CA
5	What does CSMA/CA stand for?	Carrier Sense Multiple Access with Collision Avoidance
6	[...] happens when a wireless signal passes through a material and is converted into heat, weakening the original signal.	Absorption
7	[...] happens when a signal bounces off of a material, for example metal.	Reflection
8	[...] happens when a wave is bent when entering a medium where the signal travels at a different speed.	Refraction
9	[...] happens when a wave encounters an obstacle and travels around it.	Diffraction
10	[...] happens when a material causes a signal to scatter in all directions.	Scattering
11	[...] is the maximum strength of the electric and magnetic fields of a wave.	Amplitude
12	[...] measures the number of up/down cycles per a given unit of time.	Frequency
13	The most common measurement of frequency is [...].	hertz
14	What does Hz stand for?	Hertz
15	What does kHz stand for?	Kilohertz
16	What does MHz stand for?	Megahertz
17	What does GHz stand for?	Gigahertz
18	What does THz stand for?	Terahertz
19	The term [...] refers to the amount of time for one cycle of a wave.	Period
20	Which two main bands are used for wireless LAN communications?	2.4 GHz 5 GHz
21	The Wi-Fi [...] band typically provides further reach.	2.4 GHz
22	The Wi-Fi [...] band consists of overlapping channels.	2.4 GHz
23	The Wi-Fi [...] band consists of non-overlapping channels.	5 GHz
24	To avoid overlapping frequencies, it is recommended to use channels [...], [...], and [...] of the 2.4 GHz band.	1, 6, and 11
25	802.11 max data rate:	2 Mbps
26	802.11b max data rate:	11 Mbps
27	802.11a max data rate:	54 Mbps
28	802.11g max data rate:	54 Mbps
29	802.11n max data rate:	600 Mbps
30	802.11ac max data rate:	6.93 Gbps
31	802.11ax max data rate:	4*802.11ac (4*6.93 Gbps)
32	802.11 (original) frequency/frequencies:	2.4 GHz
33	802.11b frequency/frequencies:	2.4 GHz
34	802.11a frequency/frequencies:	5 GHz
35	802.11g frequency/frequencies:	2.4 GHz
36	802.11n frequency/frequencies:	2.4 / 5 GHz
37	802.11ac frequency/frequencies:	5 GHz
38	802.11ax frequency/frequencies:	2.4 / 5 / 6 GHz
39	802.11n is aka Wi-Fi [...]	4
40	802.11ac is aka Wi-Fi [...]	5
41	802.11ax is aka Wi-Fi [...]	6
42	[...] is aka Wi-Fi 4.	802.11n
43	[...] is aka Wi-Fi 5.	802.11ac
44	[...] is aka Wi-Fi 6.	802.11ax
45	What does SSID stand for?	Service Set Identifier
46	802.11 defines three types of service sets:	Independent Infrastructure Mesh
47	The [...] is a human-readable name which identifies the service set.	SSID
48	Does an SSID have to be unique?	No
49	What does IBSS stand for?	Independent Basic Service Set

#	Question / Front	Answer / Back
50	What does AP stand for?	Access Point
51	An IBSS is also called an [...] wireless network.	ad hoc
52	What does BSS stand for?	Basic Service Set
53	What does BSSID stand for?	Basic Service Set Identifier
54	What does BSA stand for?	Basic Service Area
55	The area around an AP where its signal is usable is called a [...].	BSA (Basic Service Area)
56	The [...] is the MAC address of the AP's radio.	BSSID
57	In a BSS, can clients communicate directly with each other?	No, they must communicate via the AP
58	Wireless devices that have associated with a BSS are called clients or [...].	stations
59	What does ESS stand for?	Extended Service Set
60	When a client passes between APs in an ESS, it is called [...].	roaming
61	What does MBSS stand for?	Mesh Basic Service Set
62	In an MBSS, what does RAP stand for?	Root Access Point
63	In an MBSS, what does MAP stand for?	Mesh Access Point
64	In an MBSS, which AP is connected to the wired network?	RAP (Root Access Point)
65	In 802.11, the upstream wired network is called the [...].	DS (Distribution System)
66	An AP in [...] mode can be used to extend the range of a BSS.	repeater
67	A [...] operates as a wireless client of another AP, and can be used to connect wired devices to the wireless network.	workgroup bridge (WGB)
68	What does WGB stand for?	Workgroup bridge
69	An AP operating as an [...] can be used to connect networks over long distances without a physical cable connecting them.	outdoor bridge

Day 56 Flashcards - Wireless Architectures

57 flashcards in this section

#	Question / Front	Answer / Back
1	802.11 Frame 'Frame Control' field length:	2 bytes
2	802.11 Frame 'Duration/ID' field length:	2 bytes
3	802.11 Frame 'Address 1/2/3/4' field length:	6 bytes (per address)
4	802.11 Frame 'Sequence Control' field length:	2 bytes
5	802.11 Frame 'QoS Control' field length:	2 bytes
6	802.11 Frame 'HT Control' field length:	4 bytes
7	802.11 Frame 'FCS' field length:	4 bytes
8	There are three 802.11 connection states:1: [...] 2: [...] 3: [...]	1: Not Authenticated, not Associated.2: Authenticated, not Associated.3: Authenticated and Associated.
9	Which two messages are used for 802.11 active scanning?	Probe requestProbe response
10	Which messages does a wireless station listen for when doing 802.11 passive scanning?	Beacon
11	What are the three 802.11 message types?	ManagementControl Data
12	What 802.11 message type?Beacon	Management
13	What 802.11 message type?Probe request	Management
14	What 802.11 message type?Probe response	Management
15	What 802.11 message type?Authentication	Management
16	What 802.11 message type?Association request	Management
17	What 802.11 message type?Association response	Management
18	What 802.11 message type?RTS	Control
19	What 802.11 message type?CTS	Control
20	What 802.11 message type?ACK	Control
21	What are the three main wireless AP deployment methods? (three architectures)	AutonomousLightweight Cloud-based
22	[...] APs are self-contained systems that don't rely on a WLC.	Autonomous
23	Autonomous APs connect to a switch via [access/trunk] ports.	trunk
24	The use of lightweight APs and WLCs is also known as [...] architecture.	split-MAC
25	What does WLC stand for?	Wireless LAN Controller
26	What does CAPWAP stand for?	Control and Provisioning of Wireless Access Points

#	Question / Front	Answer / Back
27	What does LWAPP stand for?	Lightweight Access Point Protocol
28	CAPWAP [...] tunnels use UDP port 5246.	Control
29	CAPWAP Control tunnels use [...] port 5246.	UDP
30	CAPWAP Control tunnels use UDP port [...].	5246
31	CAPWAP [...] tunnels use UDP port 5247.	Data
32	CAPWAP Data tunnels use [...] port 5247.	UDP
33	CAPWAP Data tunnels use UDP port [...].	5247
34	How many CAPWAP tunnels does a lightweight AP form with the WLC?	2 (control and data)
35	The CAPWAP [...] tunnel is encrypted by default.	control
36	The CAPWAP [...] tunnel isn't encrypted by default.	data
37	The CAPWAP control tunnel [is/isn't] encrypted by default.	is
38	The CAPWAP data tunnel [is/isn't] encrypted by default.	isn't
39	Lightweight APs connect to a switch via [access/trunk] ports.	access
40	Which feature allows a lightweight AP to locally switch traffic if the tunnels to the WLC go down?	FlexConnect
41	What is the default operating mode of a lightweight AP, in which it offers a BSS for clients?	Local
42	Which lightweight AP operating mode is dedicated to capturing 802.11 frames and sending them to a device running software such as Wireshark?	Sniffer
43	Which lightweight AP operating mode is dedicated to receiving 802.11 frames to detect rogue devices?	Monitor
44	Which lightweight AP operating mode is dedicated to listening to traffic on the wired network to detect rogue devices? (it does not use its radio)	Rogue Detector
45	Which lightweight AP operating mode is dedicated to RF spectrum analysis on all channels?	SE-Connect
46	Which lightweight AP operating mode can form a dedicated bridge between sites?	Bridge/Mesh(a mesh can be formed between the APs)
47	Which lightweight AP operating mode adds FlexConnect functionality to the Bridge/Mesh mode?	Flex plus Bridge
48	Cisco [...] is a popular cloud-based AP solution.	Meraki
49	[...] AP architecture is between autonomous AP and split-MAC architecture in terms of functionality.	Cloud-based
50	A Unified WLC can support about [...] APs.	6000
51	A Cloud-based WLC can support about [...] APs.	3000
52	An Embedded WLC can support about [...] APs.	200
53	A Mobility Express WLC can support about [...] APs.	100
54	A [...] WLC is a hardware appliance in a central location of the network.	Unified
55	A [...] WLC is a VM running on a server, usually in a private cloud in a data center.	Cloud-based
56	An [...] WLC is integrated within a switch.	Embedded
57	A [...] WLC is integrated within an AP.	Mobility Express

Day 57 Flashcards - Wireless Security

43 flashcards in this section

#	Question / Front	Answer / Back
1	A [...] is used by a wireless AP to encrypt traffic that it wants to send to all of its clients.	group key
2	What does MIC stand for?	Message Integrity Check
3	What does WEP stand for?	Wired Equivalent Privacy
4	What does EAP stand for?	Extensible Authentication Protocol
5	What does LEAP stand for?	Lightweight EAP
6	What does EAP-FAST stand for?	EAP Flexible Authentication via Secure Tunneling
7	What does PEAP stand for?	Protected EAP
8	What does EAP-TLS stand for?	EAP Transport Layer Security
9	When using [...] authentication, all authentication requests are accepted.	open
10	WEP uses the [...] algorithm for encryption.	RC4
11	WEP keys can be [...] or [...] in length.	40 bits (+24 = 64) or 104 bits (+24 = 128)
12	EAP is integrated with [...], which provides port-based network access control.	802.1X
13	Which 802.1X entity? The device that wants to connect to the network.	Supplicant

#	Question / Front	Answer / Back
14	Which 802.1X entity? The device that provides access to the network.	Authenticator
15	Which 802.1X entity? The device that receives client credentials and permits/denies access.	Authentication Server (AS)
16	LEAP provides mutual authentication by exchanging [...]	challenge phrases
17	LEAP uses [...] WEP keys that change frequently.	dynamic
18	In EAP-FAST, a [...] is passed from the server to the client.	PAC (Protected Access Credential)
19	Three phases of EAP-FAST: 1: [...] 2: Secure TLS tunnel is established 3: Client is authenticated via the TLS tunnel	PAC is passed from server to client
20	Three phases of EAP-FAST: 1: PAC is passed from server to client 2: [...] 3: Client is authenticated via the TLS tunnel	Secure TLS tunnel is established
21	Three phases of EAP-FAST: 1: PAC is passed from server to client 2: Secure TLS tunnel is established 3: [...]	Client is authenticated via the TLS tunnel
22	In PEAP, the server is authenticated via a [...].	certificate
23	What does MS-CHAP stand for?	Microsoft Challenge-Handshake Authentication Protocol
24	In PEAP, which device/devices require a certificate?	AS only
25	In EAP-TLS, which device/devices require a certificate?	AS and supplicant
26	What does TKIP stand for?	Temporal Key Integrity Protocol
27	What does CCMP stand for?	Counter/CBC-MAC Protocol
28	What does GCMP stand for?	Galois/Counter Mode Protocol
29	Which encryption protocol was developed as an improvement upon WEP?	TKIP
30	What does CCMP use for encryption?	AES counter mode
31	What does CCMP use as a MIC?	CBC-MAC
32	What does CBC-MAC stand for?	Cipher Block Chaining Message Authentication Code
33	What does GCMP use for encryption?	AES counter mode
34	What does GCMP use for MIC?	GMAC
35	What does GMAC stand for?	Galois Message Authentication Code
36	What are the two WPA authentication modes?	Personal, Enterprise
37	Which WPA authentication mode uses a PSK?	Personal mode
38	Which WPA authentication mode uses 802.1X/EAP?	Enterprise mode
39	WPA uses [...] for encryption/MIC.	TKIP
40	WPA2 uses [...] for encryption/MIC.	CCMP
41	WPA3 uses [...] for encryption/MIC.	GCMP
42	WPA3 [...] protects the four-way handshake when using personal mode authentication.	SAE (Simultaneous Authentication of Equals)
43	WPA3 [...] prevents data from being decrypted after it has been transmitted over the air.	forward secrecy

Day 58 Flashcards - Wireless Configuration

34 flashcards in this section

#	Question / Front	Answer / Back
1	WLCs only support [...] LAG.	static *PAGP and LACP aren't supported
2	DHCP option [...] can be used to tell APs the IP address of their WLC.	43
3	In the context of WLCs: Port = [physical/logical] Interface = [physical/logical]	Port = physical Interface = logical
4	The WLC [...] port is used for out-of-band management.	service
5	The WLC service port must connect to a switch [access/trunk] port	access
6	WLC [...] ports are the standard network ports that connect to the switched network and are used for data traffic.	distribution system
7	The WLC [...] port is used to connect to another WLC to form an HA pair.	redundancy
8	CAPWAP tunnels are formed to/from the WLC's [...] interface.	management

#	Question / Front	Answer / Back
9	The WLC [...] interface can be used to connect to and manage the standby WLC of an HA pair.	redundancy management
10	The WLC [...] interface is used when communicating with wireless clients to relay DHCP requests, perform web authentication, etc.	virtual
11	The WLC [...] interface is bound to the service port and used for out-of-band management.	service port
12	WLC [...] interfaces are used to map WLANs to VLANs.	dynamic
13	WPA+WPA2 is a Layer [...] security mechanism.	2
14	802.1X is a Layer [...] security mechanism.	2
15	Static WEP is a Layer [...] security mechanism.	2
16	Static WEP + 802.1X is a Layer [...] security mechanism.	2
17	CKIP is a Layer [...] security mechanism.	2 (CKIP = Cisco Key Integrity Protocol)
18	None + EAP Passthrough is a Layer [...] security mechanism.	2
19	What formats can a WLAN PSK be configured in?	ASCII / HEX
20	WLAN PSKs in ASCII format must be at least [...] characters in length.	8
21	Web Authentication is a Layer [...] security mechanism.	3
22	Web Passthrough is a Layer [...] security mechanism.	3
23	Conditional Web Redirect is a Layer [...] security mechanism.	3
24	Splash Page Web Redirect is a Layer [...] security mechanism.	3
25	[...] is the default QoS setting of a WLAN.	Silver
26	WLAN Platinum QoS = [...]	voice
27	WLAN Gold QoS = [...]	video
28	WLAN Silver QoS = [...]	best effort
29	WLAN Bronze QoS = [...]	background
30	WLAN [...] QoS = voice	Platinum
31	WLAN [...] QoS = video	Gold
32	WLAN [...] QoS = best effort	Silver
33	WLAN [...] QoS = background	Bronze
34	[...] ACLs can be used to limit management traffic destined for the WLC itself.	CPU

Day 59 (Part 1) Flashcards - Network Automation

22 flashcards in this section

#	Question / Front	Answer / Back
1	What are the three logical planes of network functions?	Management plane Control plane Data plane
2	Tasks involved in forwarding user traffic from one interface to another are part of the [...] plane.	data
3	Functions that control the actions of the data plane are part of the [...] plane.	control
4	Functions that are used to manage devices are part of the [...] plane.	management
5	The Data plane is also known as the [...] plane.	Forwarding
6	What does ASIC stand for?	Application-Specific Integrated Circuit
7	What does TCAM stand for?	Ternary Content-Addressable Memory
8	The MAC address table is also known as the [...] table.	CAM
9	What does SDN stand for?	Software-Defined Networking
10	What does SDA stand for?	Software-Defined Architecture
11	What does API stand for?	Application Programming Interface
12	SDN is an approach to networking that centralizes the [...] plane into an application called a controller.	control
13	What does SBI stand for?	Southbound Interface
14	What does NBI stand for?	Northbound Interface
15	OpenFlow is a [...]bound interface.	south
16	OpFlex is a [...]bound interface.	south
17	onePK is a [...]bound interface.	south
18	NETCONF is a [...]bound interface.	south
19	The [...]bound interface is used for communications between the SDN controller and the network devices it controls.	South

#	Question / Front	Answer / Back
20	The [...]bound interface is used for communications between the SDN controller and applications.	North
21	What does REST stand for?	Representational State Transfer
22	REST APIs are used for [...]bound interfaces.	North

Day 59 (part 2) Flashcards - AI & Machine Learning

16 flashcards in this section

#	Question / Front	Answer / Back
1	What does AI stand for?	artificial intelligence
2	[...] uses computers to simulate intelligence, allowing them to exhibit behaviors typically associated with humans.	AI
3	What does ML stand for?	Machine Learning
4	[...] is a subset of AI that enables computers to learn from data and improve without explicit programming.	ML
5	ML: [...] learning trains the model on a labeled dataset.	Supervised
6	ML: [...] learning trains the model on an unlabeled dataset.	Unsupervised
7	ML: [...] learning trains the model by rewarding or penalizing its actions in a given environment.	Reinforcement
8	ML: [...] learning uses artificial neural networks to process and learn from large and complex datasets.	Deep
9	[...] AI analyzes historical data to predict future outcomes or trends.	Predictive
10	[...] AI learns patterns from existing data to create new content.	Generative
11	Catalyst Center [...] uses AI to establish the baseline behavior of the network and provide recommendations for optimizing the network.	AI Network Analytics
12	Catalyst Center [...] uses AI to perform root-cause analysis and take corrective actions when network issues arise.	Machine Reasoning Engine (MRE)
13	What does Catalyst Center MRE stand for?	Machine Reasoning Engine
14	Catalyst Center [...] uses AI to automatically identify and classify devices connected to the network.	AI Endpoint Analytics
15	Catalyst Center [...] uses AI to optimize wireless network performance by dynamically adjusting radio settings.	AI-enhanced Radio Resource Management (RRM)
16	What does RRM stand for?	Radio Resource Management

Day 60 Flashcards - JSON, XML, & YAML

26 flashcards in this section

#	Question / Front	Answer / Back
1	What does JSON stand for?	JavaScript Object Notation
2	What does XML stand for?	Extensible Markup Language
3	What does YAML stand for?	YAML Ain't Markup Language
4	Which data serialization languages are often used by REST APIs?	JSON, XML
5	In JSON, whitespace [is/isn't] significant.	isn't
6	In XML, whitespace [is/isn't] significant.	isn't
7	In YAML, whitespace [is/isn't] significant.	is
8	What are the four JSON 'primitive' data types?	-string -number -boolean -null
9	What are the two JSON 'structured' data types?	-object -array
10	What kind of JSON data type is the value of this key-value pair? "key": "value"	string
11	What kind of JSON data type is the value of this key-value pair? "key": 5	number
12	What kind of JSON data type is the value of this key-value pair? "key": false	boolean
13	What kind of JSON data type is the value of this key-value pair? "key": null	
14	What kind of JSON data type is the value of this key-value pair? "key": ["yes", "no"]	array

#	Question / Front	Answer / Back
15	What kind of JSON data type is the value of this key-value pair? "key":{"interface":"gigabitethernet1/1"}	object
16	A JSON [...] is a text value.	string
17	A JSON [...] is a numeric value.	number
18	A JSON [...] can be true or false.	boolean
19	A JSON [...] data type represents the intentional absence of any value.	
20	A JSON [...] is a list of key-value pairs.	object
21	A JSON [...] is a series of values.	array
22	[serialization language] uses the <key>value</key> format.	XML
23	YAML files start with [...].	---
24	[serialization language] files start with ---.	YAML
25	In YAML, [...] indicates a list.	-
26	YAML key-value pairs are represented as [...].	key:value

Day 61 (Part 1) Flashcards - REST APIs

38 flashcards in this section

#	Question / Front	Answer / Back
1	What does CRUD stand for?	Create, Read, Update, Delete
2	CRUD [...] = HTTP POST	Create
3	CRUD Create = HTTP [...]	POST
4	CRUD [...] = HTTP GET	Read
5	CRUD Read = HTTP [...]	GET
6	CRUD [...] = HTTP PUT, PATCH	Update
7	CRUD Update = HTTP [...]	PUT, PATCH
8	CRUD [...] = HTTP DELETE	Delete
9	CRUD Delete = HTTP [...]	DELETE
10	What does URI stand for?	Uniform Resource Identifier
11	Which part of the following URI is the scheme? https://sandboxdnac.cisco.com/dna/intent/api/v1/network-device	https
12	Which part of the following URI is the authority? https://sandboxdnac.cisco.com/dna/intent/api/v1/network-device	sandboxdnac.cisco.com
13	Which part of the following URI is the path? https://sandboxdnac.cisco.com/dna/intent/api/v1/network-device	/dna/intent/api/v1/network-device
14	HTTP Response class [...] = informational	1xx
15	HTTP Response class 1xx = [...]	informational
16	HTTP Response class [...] = successful	2xx
17	HTTP Response class 2xx = [...]	successful
18	HTTP Response class [...] = redirection	3xx
19	HTTP Response class 3xx = [...]	redirection
20	HTTP Response class [...] = client error	4xx
21	HTTP Response class 4xx = [...]	client error
22	HTTP Response class [...] = server error	5xx
23	HTTP Response class 5xx = [...]	server error
24	HTTP Response code [...] = OK	200
25	HTTP Response code 200 = [...]	OK
26	HTTP Response code [...] = Processing	102
27	HTTP Response code 102 = [...]	Processing
28	HTTP Response code [...] = Created	201
29	HTTP Response code 201 = [...]	Created
30	HTTP Response code [...] = Moved Permanently	301
31	HTTP Response code 301 = [...]	Moved Permanently
32	HTTP Response code [...] = Unauthorized	401
33	HTTP Response code 401 = [...]	Unauthorized
34	HTTP Response code [...] = Not Found	404
35	HTTP Response code 404 = [...]	Not Found
36	HTTP Response code [...] = Internal Server Error	500

#	Question / Front	Answer / Back
37	HTTP Response code 500 = [...]	Internal Server Error
38	REST APIs must be [stateful/stateless].	stateless

Day 61 (Part 2) Flashcards - REST Authentication

12 flashcards in this section

#	Question / Front	Answer / Back
1	Which form of REST API authentication uses a username/password combination?	Basic authentication
2	In REST API Basic authentication, credentials are encoded in [format].	Base64
3	Is Base64 encoding secure?	No. Encoding is easily reversible; it is not encryption.
4	REST API Bearer authentication uses a [...] for authentication.	token (also called a bearer token)
5	In REST API Basic authentication, in which part of the message are the credentials included?	In the HTTP Authorization header
6	In REST API Bearer authentication, in which part of the message are the credentials included?	In the HTTP Authorization header
7	In REST API Bearer authentication, are the bearer tokens valid indefinitely?	No; they expire after a set period of time
8	API key authentication uses a [...] key issued by the API provider.	static (non-expiring)
9	In API key authentication, in which part of the message should the client specify the key?	The HTTP Authorization header (recommended). *Other options: URL, cookie
10	REST API [...] authentication provides access delegation.	OAuth 2.0
11	What are the four entities in OAuth 2.0?	Resource owner Client app Auth server Resource server
12	OAuth 2.0: A [...] can be used to renew access tokens without user reauthentication.	refresh token

Day 62 Flashcards - SDN

19 flashcards in this section

#	Question / Front	Answer / Back
1	The [...] layer of SDN architecture contains scripts and applications that interact with the SDN controller.	Application
2	The [...] layer of SDN architecture contains the SDN controller.	Control
3	The [...] layer of SDN architecture contains the network devices.	Infrastructure
4	What does Cisco DNA stand for?	Digital Network Architecture
5	What does Cisco ACI stand for?	Application-Centric Infrastructure
6	[...] is Cisco's SDN solution for automating campus LANs.	SD-Access
7	[...] is Cisco's SDN solution for automating data center networks.	ACI
8	[...] is Cisco's SDN solution for automating WANs.	SD-WAN
9	In SDN, the [...] is the physical network of devices and connections.	underlay
10	In SDN, the [...] is the virtual network built on top of the physical network.	overlay
11	In SDN, the [...] is the combination of the overlay and underlay.	fabric
12	What does VXLAN stand for?	Virtual Extensible LAN
13	What does LISP stand for?	Locator ID Separation Protocol
14	What are the three switch types in Cisco SD-Access?	-Edge node -Border node -Control node
15	What does CTS stand for?	Cisco TrustSec
16	What does IBN stand for?	Intent-Based Networking
17	[protocol] provides the control plane of Cisco SD-Access.	LISP
18	[protocol] provides the data plane of Cisco SD-Access.	VXLAN
19	[protocol] provides policy control in Cisco SD-Access.	Cisco TrustSec

Day 63 Flashcards - Ansible, Puppet, Chef

27 flashcards in this section

#	Question / Front	Answer / Back
1	Ansible is written in [language].	Python

#	Question / Front	Answer / Back
2	Ansible is [agent-based/agentless].	agentless
3	Puppet is [agent-based/agentless].	agent-based *with an external agent, the managed devices can be agentless
4	Chef is [agent-based/agentless].	agent-based
5	Puppet is written in [language].	Ruby
6	Chef is written in [language].	Ruby
7	Ansible uses [protocol] to connect to devices.	SSH
8	Chef uses [protocol] to connect to devices.	HTTPS
9	Puppet uses [protocol] to connect to devices.	HTTPS
10	Ansible uses a [push/pull] model.	push
11	Puppet uses a [push/pull] model.	pull
12	Chef uses a [push/pull] model.	pull
13	Ansible [...] files define the actions to be taken.	playbook
14	Puppet [...] files define the actions to be taken.	manifest
15	Chef [...] files define the actions to be taken.	recipe / run-list
16	[...] clients use TCP port 8140 to communicate with the server.	Puppet
17	Puppet clients use [...] port 8140 to communicate with the server.	TCP
18	Puppet clients use TCP port [...] to communicate with the server.	8140
19	A Puppet server is called a [...].	Puppet master
20	The Ansible server is called the [...].	Control node
21	[Ansible/Puppet/Chef] files are written in YAML.	Ansible *templates use Jinja2
22	[Ansible/Puppet/Chef] files are written in a proprietary language.	Puppet / Chef
23	Puppet files are written in [language].	a proprietary language (known as Puppet DSL)
24	Chef files are written in [language].	a proprietary language based on Ruby.
25	[...] uses TCP port 10002 to send configurations to clients.	Chef
26	Chef uses [...] port 10002 to send configurations to clients.	TCP
27	Chef uses TCP port [...] to send configurations to clients.	10002

Day 63 (part 2) Flashcards - Terraform

27 flashcards in this section

#	Question / Front	Answer / Back
1	What does IaC stand for?	Infrastructure as Code
2	Ansible, Puppet, Chef, and Terraform are examples of [...] tools.	Infrastructure as Code (IaC)
3	Ansible is primarily a [configuration management/provisioning] tool.	configuration management
4	Puppet is primarily a [configuration management/provisioning] tool.	configuration management
5	Chef is primarily a [configuration management/provisioning] tool.	configuration management
6	Terraform is primarily a [configuration management/provisioning] tool.	provisioning
7	Configuration management tools typically use a(n) [mutable/immutable] infrastructure approach.	mutable
8	Provisioning tools typically use a(n) [mutable/immutable] infrastructure approach.	immutable
9	Ansible uses a(n) [mutable/immutable] infrastructure approach.	mutable
10	Puppet uses a(n) [mutable/immutable] infrastructure approach.	mutable
11	Chef uses a(n) [mutable/immutable] infrastructure approach.	mutable
12	Terraform uses a(n) [mutable/immutable] infrastructure approach.	immutable
13	A [...] approach defines explicit steps in a specific order to achieve the desired outcome.	procedural
14	A [...] approach defines the desired end state, and lets the tool figure out the steps needed to achieve that state.	declarative
15	Ansible uses a [procedural/declarative] approach.	procedural
16	Chef uses a [procedural/declarative] approach.	procedural
17	Terraform uses a [procedural/declarative] approach.	declarative
18	Puppet uses a [procedural/declarative] approach.	declarative
19	Terraform provisions infrastructure resources on platforms called [...].	providers
20	Terraform uses a [push/pull] model.	push

#	Question / Front	Answer / Back
21	Terraform uses an [agent-based/agentless] architecture.	agentless
22	Terraform [...] is the central Terraform software that analyzes configurations and interacts with providers.	Core
23	Terraform [...] files define the desired end state of the infrastructure.	configuration
24	The Terraform [...] file tracks the status of the deployed infrastructure.	state
25	What the the three main steps in the basic Terraform workflow?	Write, Plan, Apply
26	Terraform Core is written in [language].	Go
27	Terraform configuration files are written in [language].	HashiCorp Configuration Language (HCL)